

Module 4. SELECTED TOPICS IN GRAPH THEORY

4.1 Graph Terminology

4.2 Representation of Graphs and
Isomorphism

4.3 Shortest-Path Problems.

4.4 Connectivity

4.5 Euler and Hamilton Paths

4.6 Planar Graphs

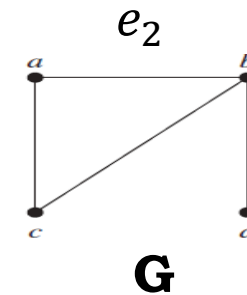
4.7 Graph Colouring

4.1 Graph Terminology

To solve the Königsberg bridge problem, in 1937, a Swiss mathematician Leonhard Euler used the notion of graph theory. A **graph theory is a** branch of mathematics concerned with networks of points (vertices or nodes) connected by lines (edges).

Graph: A *graph* $G = (V, E)$ consists of V , a nonempty set of *vertices* (or *nodes*) and E , a set of *edges*. Each edge has either one or two vertices associated with it, called its *endpoints*. An edge is said to *connect* its endpoints.

Example: In the graph G
Vertex set $V(G) = \{a, b, c, d\}$, Edge set $E(G) = \{e_1, e_2, e_3, e_4\}$

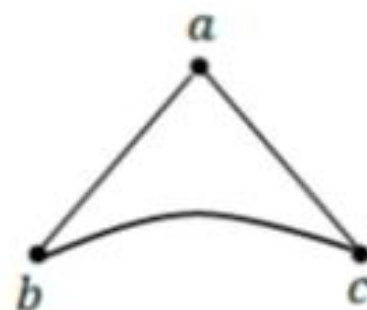


4.1.1 Basics of Graph

Remark: The set of vertices V of a graph G may be infinite. A graph with an infinite vertex set or an infinite number of edges is called an **infinite graph**, and in comparison, a graph with a finite vertex set and a finite edge set is called a **finite graph**. In this module we will consider only finite graphs.

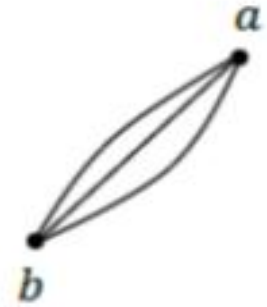
We write the edge e as $\{u, v\}$, which is the edge between the two vertices u and v . Note that, the notations $\{u, v\}$ and $\{v, u\}$ are the same.

Definition: A **simple graph** is a graph in which each edge connects two different vertices and where no two edges connect the same pair of vertices.



In another word, If there is an edge $\{u, v\}$ between the vertices u and v in a simple graph G , then there is no other edge is associated with these two vertices. Which means, each pair of vertices has exactly one edge is associated with.

Definition: A **multiple edge** is more than one edge connecting the same pair of vertices in a graph. If there are m different multiple edges associated with same unordered pair of vertices $\{u, v\}$, then we say that the edge $\{u, v\}$ has multiplicity m .



Definition: A **multigraph** is a graph that has multiple edges.

Definition: A **loop** is an edge that connects a vertex to itself.



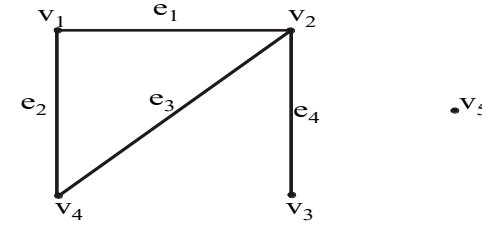
Definition: A **pseudograph** is a graph contains loops and possibly multiple edges



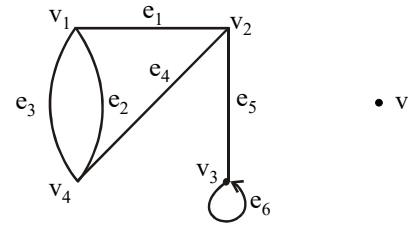
Definition: When the edge starts at the vertex u and ends at vertex v is called **directed edge**, and denoted by (u, v) .



Simple Graph: A simple graph is a graph that does not have any loop or parallel edges. For example, the graph H is a simple graph with vertex set $V(H) = \{v_1, v_2, v_3, v_4, v_5\}$ and edge set $E(H) = \{e_1, e_2, e_3, e_4\}$



Example: The graph G with $V(G) = \{v_1, v_2, v_3, v_4, v_5\}$ and $E(G) = \{e_1, e_2, e_3, e_4, e_5, e_6\}$

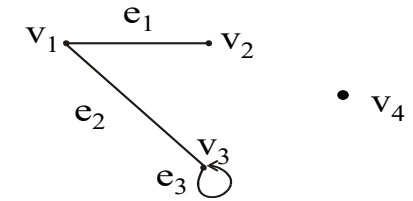


Note:

1. An edge connects either one or two vertices called its **endpoints** (In graph G: edge e_1 connects vertices v_1 and v_2 described as $\{v_1, v_2\}$, i.e., v_1 and v_2 are the endpoints of an edge e_1).
2. An edge with just one endpoint is called a **loop**. Thus a loop is an edge that connects a vertex to itself (e.g., edge e_6 makes a loop as it has only one endpoint v_3).
3. Two vertices that are connected by an edge are called **adjacent**; and a vertex that is an endpoint of a loop is said to be adjacent to itself.
4. An edge is said to be **incident** on each of its endpoints (i.e. e_1 is incident on v_1 and v_2).

5. A vertex on which no edges are incident is called **isolated** (e.g., v_5)
6. Two distinct edges with the same set of end points are said to be **parallel** (i.e. e_2 & e_3).

Example: Define the following graph formally by specifying its vertex set, its edge set, and a table giving the edge endpoint function.



Solution:

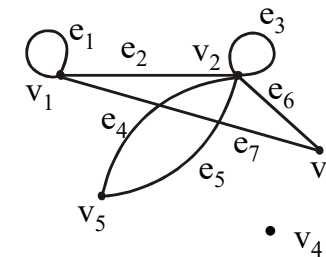
Vertex Set = $\{v_1, v_2, v_3, v_4\}$

Edge Set = $\{e_1, e_2, e_3\}$

Edge - endpoint function	
Edge	Endpoint
e_1	$\{v_1, v_2\}$
e_2	$\{v_1, v_3\}$
e_3	$\{v_3\}$

Example: For the graph shown below:

- (i) find all edges that are incident on v_1 ;
- (ii) find all vertices that are adjacent to v_3 ;
- (iii) find all loops;
- (iv) find all parallel edges;
- (v) find all isolated vertices;

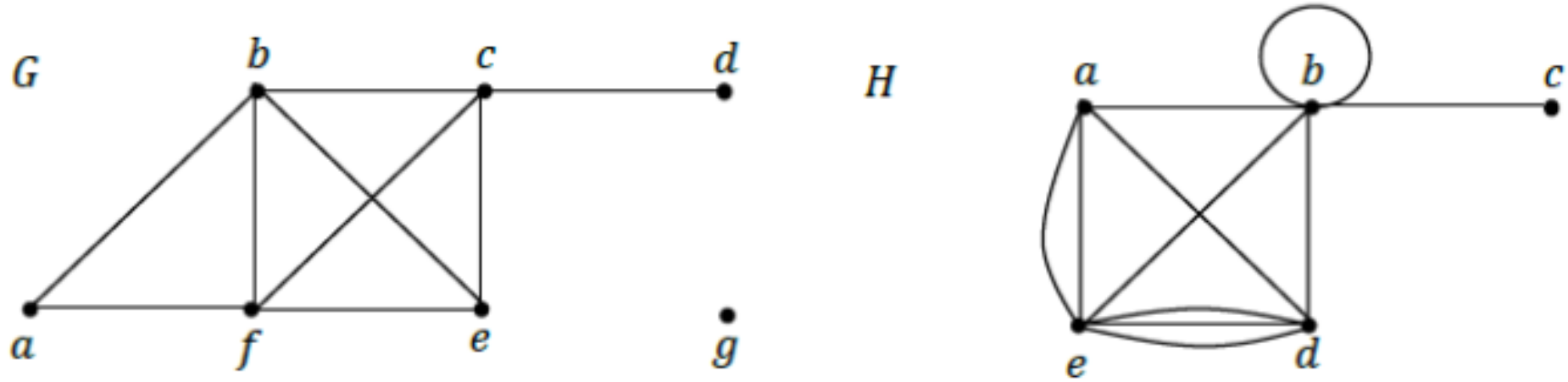


Solution: (i) v_1 is incident with edges e_1 , e_2 and e_7 .
(ii) vertices adjacent to v_3 are v_1 and v_2 .
(iii) loops are e_1 and e_3 .
(iv) only edges e_4 and e_5 are parallel.
(v) The only isolated vertex is v_4 in this Graph.

Note: Two vertices u and v in an undirected graph G are called *adjacent* (or *neighbors*) in G if u and v are endpoints of an edge e of G . Such an edge e is called *incident with* the vertices u and v and e is said to *connect* u and v .

Degree of vertex: The *degree of a vertex in an undirected graph* is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex. The degree of the vertex v is denoted by $\deg(v)$.

Example : What are the degrees of the vertices in the graphs G and H displayed in Figure below?



Solution: In G , $\deg(a) = 2$, $\deg(b) = \deg(c) = \deg(f) = 4$, $\deg(d) = 1$, $\deg(e) = 3$, and $\deg(g) = 0$. In H , $\deg(a) = 4$, $\deg(b) = \deg(e) = 6$, $\deg(c) = 1$, and $\deg(d) = 5$.

Remarks:

- (i) A vertex of degree zero is called **isolated**.
- (ii) A vertex is **pendant** if and only if it has degree one.

THE HANDSHAKING THEOREM (for undirected graph):

Let $G = (V, E)$ be an undirected graph with e edges. Then

$$2e = \sum_{v \in V} \deg(v)$$

Remarks:

- (i) An undirected graph has an even number of vertices of odd degree.
- (ii) The sum of the degrees of the vertices of an undirected graph is even.

Example: How many edges are there in a graph with 10 vertices each of degree six?

Solution: The sum of the degrees of the vertices is $6 \cdot 10 = 60$, it follows that $2e = 60$ (by Handshaking theorem), where e is the number of edges. Therefore, $e = 30$.

Example: Draw a graph with the specified properties or explain why no such graph exists.

Graph with four vertices of degrees 1, 2, 3 and 3

Graph with four vertices of degrees 1, 2, 3 and 4

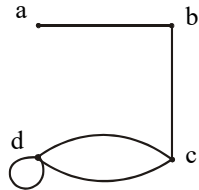
Simple graph with four vertices of degrees 1, 2, 3 and 4

Solution: (i) Total degree of graph $= 1 + 2 + 3 + 3$
 $= 9$, an odd integer.

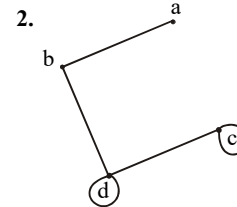
Since, the total degree of a graph is always even, hence no such graph is possible.

(ii) Since the total degree of the graph $= 1 + 2 + 3 + 4 = 10$ (even number), the graph is possible. The two graphs with four vertices of degrees 1, 2, 3 & 4 are

1.



or



Obviously, the vertices a, b, c, d have degrees 1, 2, 3, and 4 respectively.

(iii) Suppose there was a simple graph with four vertices of degrees 1, 2, 3, and 4. Then the vertex of degree 4 would have to be connected by edges to four distinct vertices other than itself because of the assumption that the graph is simple (and hence has no loop or parallel edges). This contradicts the assumption that the graph has four vertices in total.

Hence there is no simple graph with four vertices of degrees 1, 2, 3, and 4, so simple graph is not possible in this case.

Example: Suppose a graph has vertices of degrees 1, 1, 4, 4 and 6. How many edges does the graph have?

Solution: The total degree of graph = $1 + 1 + 4 + 4 + 6$
 $= 16$

Since, the total degree of graph = $2 \times (\text{Number of edges of graph})$
[By using Handshaking theorem]

$$\Rightarrow 16 = 2 \times (\text{Number of edges of graph})$$

$$\frac{16}{2} = 8$$

$\Rightarrow$ Number of edges of graph

Example: In a group of 15 people, is it possible for each person to have exactly 3 friends?

Solution: Suppose that in a group of 15 people, each person had exactly 3 friends. Then we could draw a graph representing each person by a vertex and connecting two vertices by an edge if the corresponding people were friends. But such a graph would have 15 vertices each of degree 3, for a total degree of 45 (not even), which is not possible. Hence, in a group of 15 people it is not possible for each to have exactly three friends.

In-degree and out-degree of a vertex v : In a graph with directed edges the *in-degree of a vertex v* , denoted by $\deg^-(v)$, is the number of edges with v as their terminal vertex. The *out-degree of v* , denoted by $\deg^+(v)$, is the number of edges with v as their initial vertex.

Note: that a loop at a vertex contributes 1 to both the in-degree and the out-degree of this vertex.

Example: Find the in-degree and out-degree of each vertex in the graph G with directed edges.

Solution: The *in-degrees* in G are

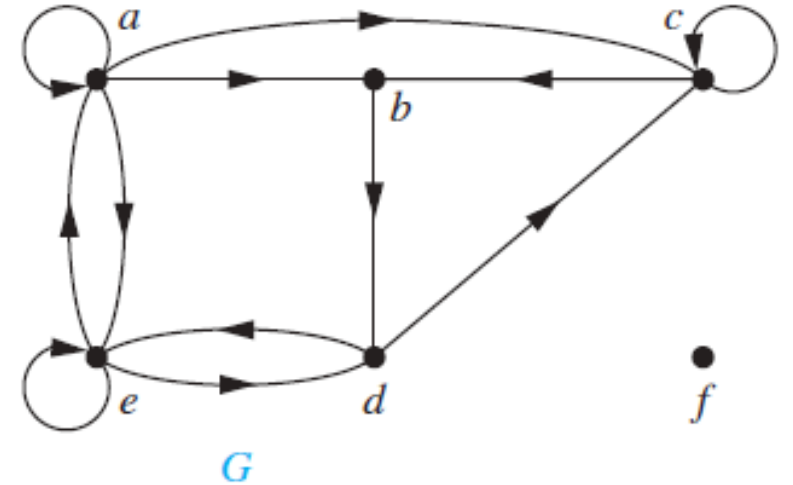
$$\deg^{-}(a) = 2, \deg^{-}(b) = 2, \deg^{-}(c) = 3,$$

$$\deg^{-}(d) = 2, \deg^{-}(e) = 3, \text{ and } \deg^{-}(f) = 0.$$

The *out-degrees* in G are

$$\deg^{+}(a) = 4, \deg^{+}(b) = 1, \deg^{+}(c) = 2,$$

$$\deg^{+}(d) = 2, \deg^{+}(e) = 3, \text{ and } \deg^{+}(f) = 0.$$



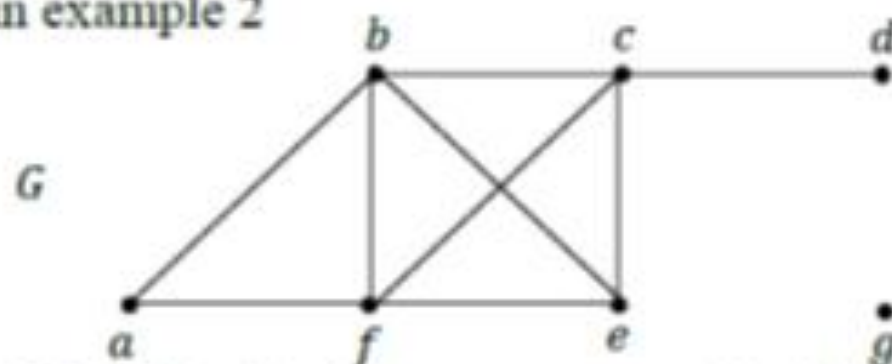
THE HANDSHAKING THEOREM (for directed graph):

Let $G = (V, E)$ be a directed graph with e edges. Then

$$|E| = \sum_{v \in V} \deg^{-}(v) = \sum_{v \in V} \deg^{+}(v)$$

Definition: The **degree sequence** of a graph is the sequence of the degrees of the vertices of the graph in a nonincreasing order.

Example : The degree sequence of the graph G in example 2 is 4,4,4,3,2,1,0



Definition: A sequence $d_1, d_2, \dots, d_n$ is called **graphic** if it is the degree sequence of a simple graph.

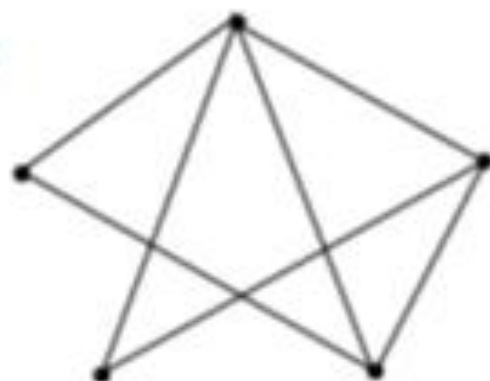
Example : Draw the graphs of the following degree sequences, and determine whether or not these sequences are graphics.

(a) 4,3,3,2,2

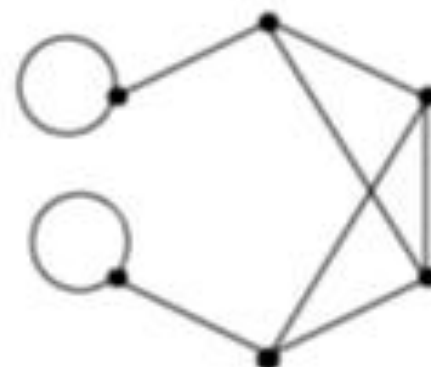
(b) 3,3,3,3,3,3

Solution:

(a) Is a graphic



(b) Is not a graphic

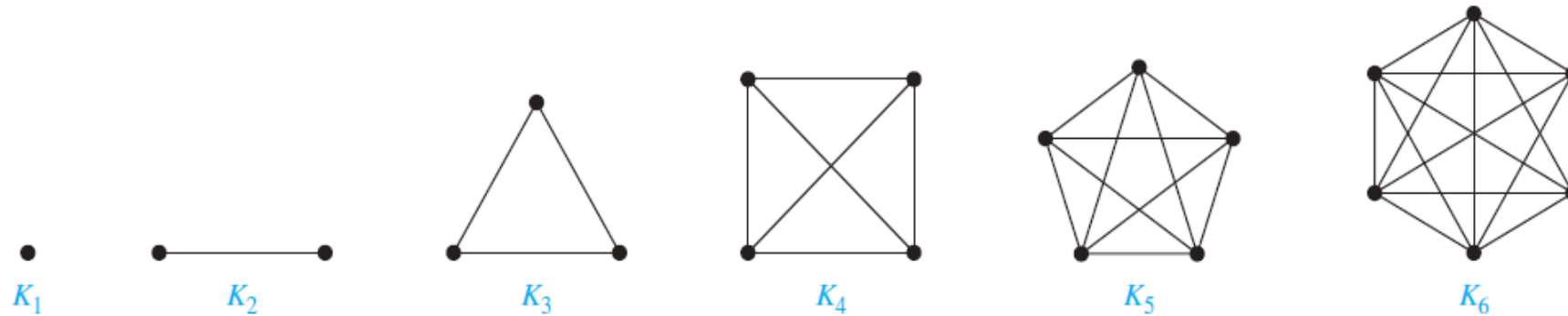


Exercise 4.1

4.1.2. Some Special Types of Simple Graphs:

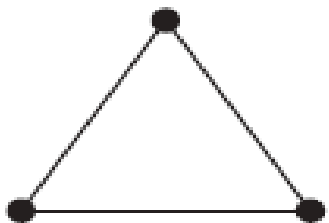
Complete Graphs:

A **complete graph on n vertices**, denoted by K_n , is a simple graph that contains exactly one edge between each pair of distinct vertices. The graphs K_n , for $n = 1, 2, 3, 4, 5, 6$, are displayed in below figure.

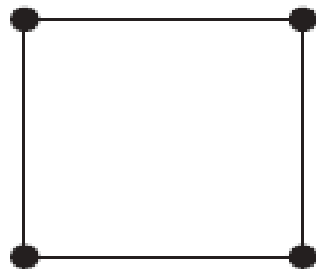


Regular Graphs:

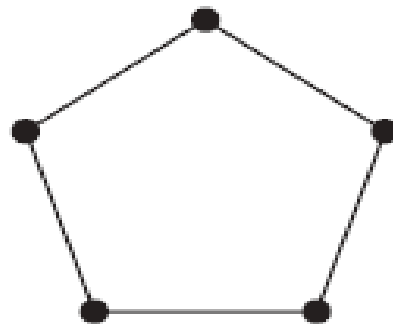
A simple graph is called **regular** if every vertex has the same degree. A regular graph is called **n –regular** if every vertex in this graph has degree n . For example, C_3, C_4, C_5, C_6 are regular because each vertices have same degree and it is equal to 2.



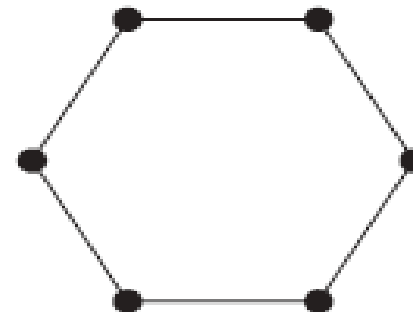
C_3



C_4



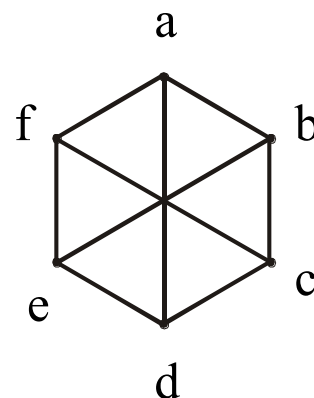
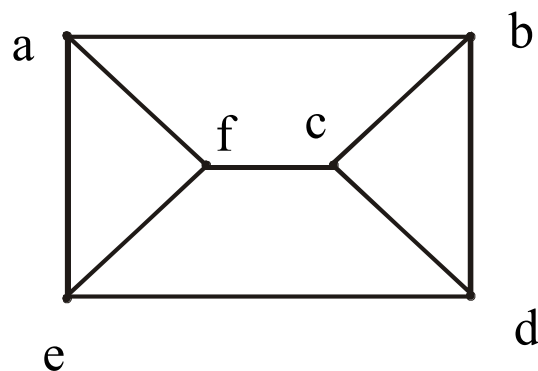
C_5



C_6

Example: Draw two 3-regular graphs with six vertices.

Solution:

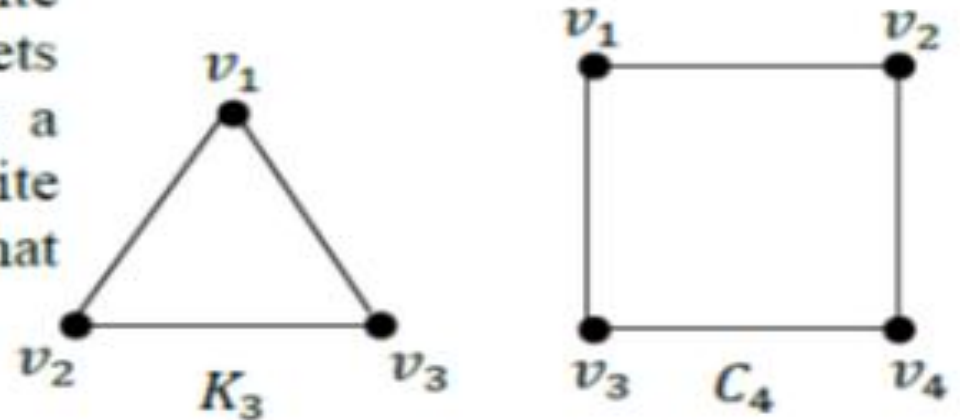


Bipartite Graphs:

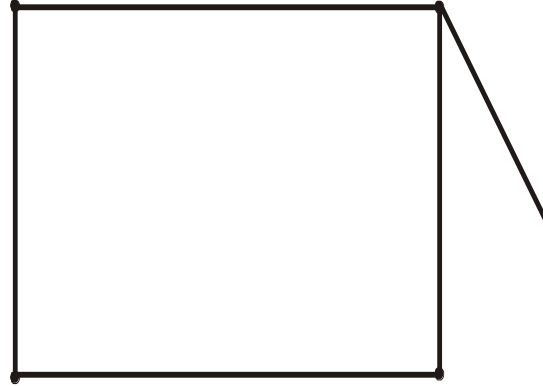
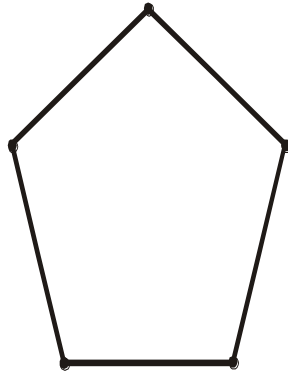
A simple graph G is called **bipartite** if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 (so that no edge in G connects either two vertices in V_1 or two vertices in V_2). When this condition holds, we call the pair (V_1, V_2) a *bipartition* of the vertex set V of G .

Example:

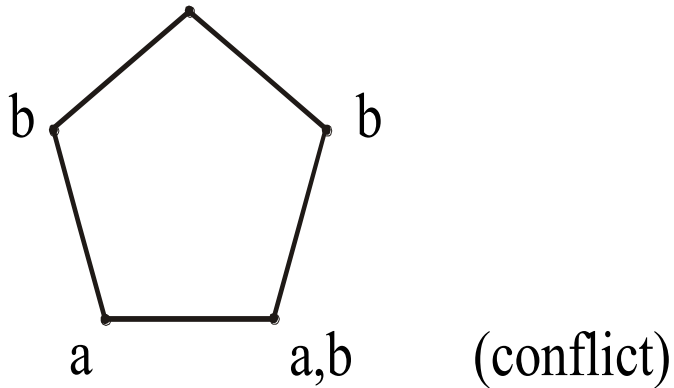
From the figure here, we can see that C_4 is bipartite because its vertex set can be partitioned into the two sets $V_1 = \{v_1, v_4\}, V_2 = \{v_2, v_3\}$ and every edge connects a vertex in V_1 and a vertex in V_2 . While K_3 is not bipartite because there are no disjoint subsets of vertices that edges do not connect two vertices in the same set.



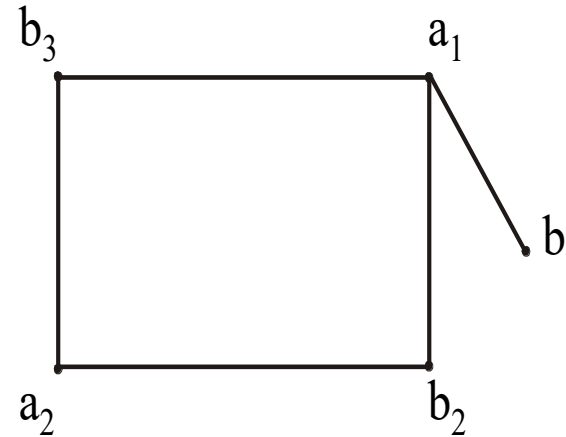
Example: Find which of the following graphs are bipartite.



Solution:



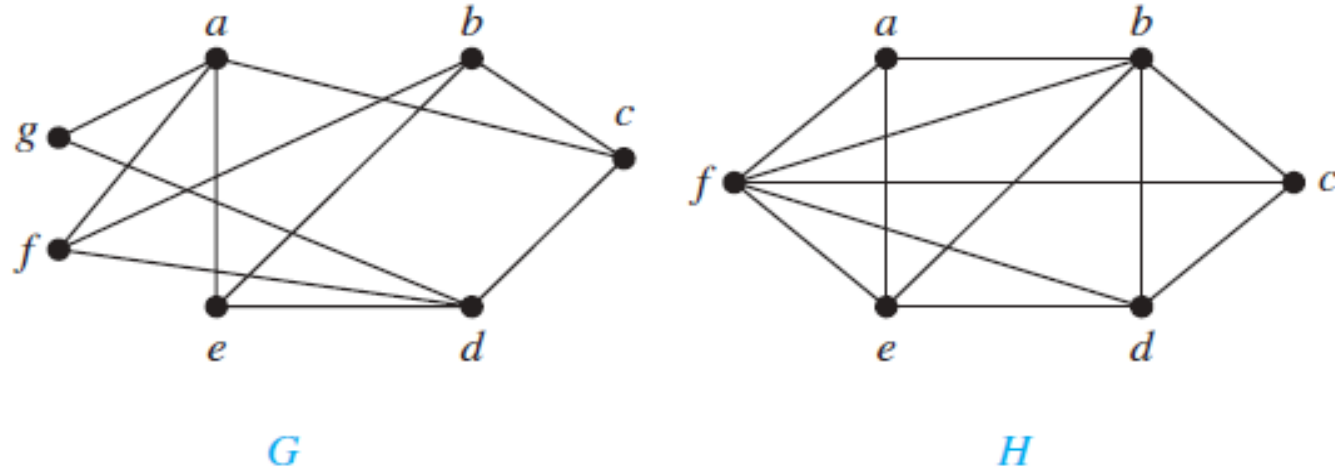
The graph is not bipartite.



$$V_1 = \{a_1, a_2\}, V_2 = \{b_1, b_2, b_3\}$$

By labeling procedure, each vertex gets a distinct label. Hence the graph is bipartite

Example: Are the graphs G and H bipartite?

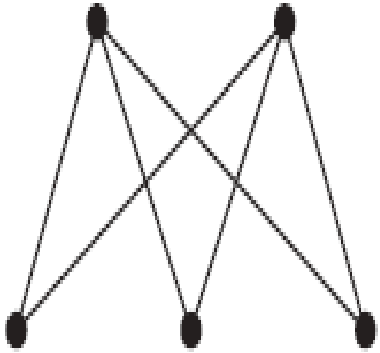


Solution: Graph G is bipartite because its vertex set is the union of two disjoint sets, $\{a, b, d\}$ and $\{c, e, f, g\}$, and each edge connects a vertex in one of these subsets to a vertex in the other subset. Graph H is not bipartite because its vertex set cannot be partitioned into two subsets so that edges do not connect two vertices from the same subset.

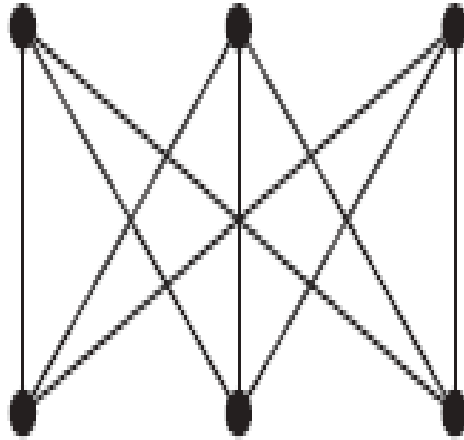
Note: A simple graph is bipartite if and only if it is possible to assign one of two different colors to each vertex of the graph so that no two adjacent vertices are assigned the same color.

Complete Bipartite Graphs: A **complete bipartite graph** $K_{m,n}$ is a graph that has its vertex set partitioned into two subsets of m and n vertices, respectively with an edge between two vertices if and only if one vertex is in the first subset and the other vertex is in the second subset.

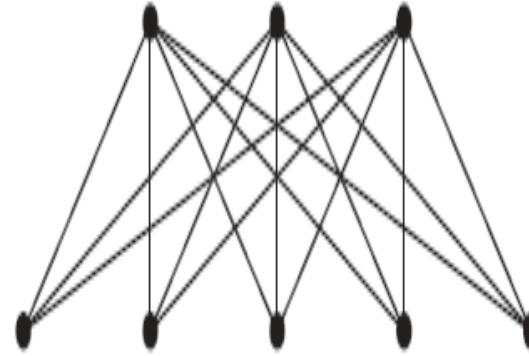
Example:



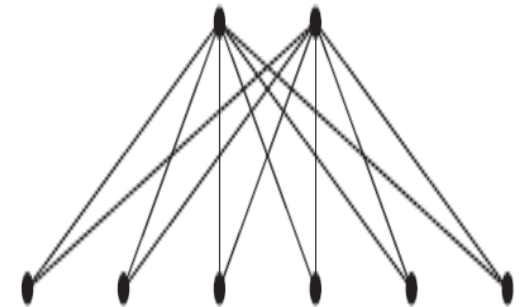
$K_{2,3}$



$K_{3,3}$



$K_{3,5}$

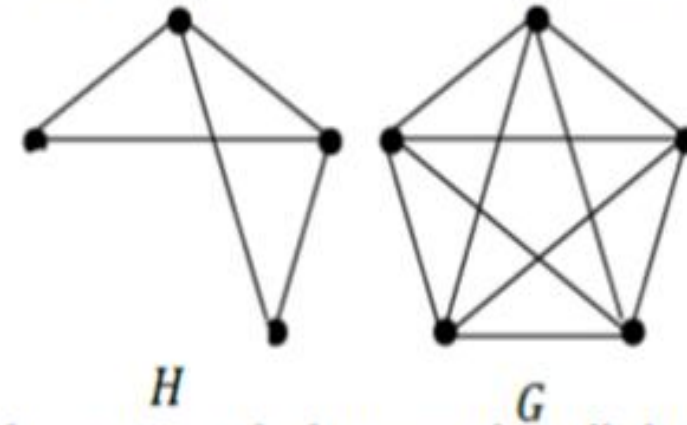


$K_{2,6}$

4.1.3. New Graphs from Old:

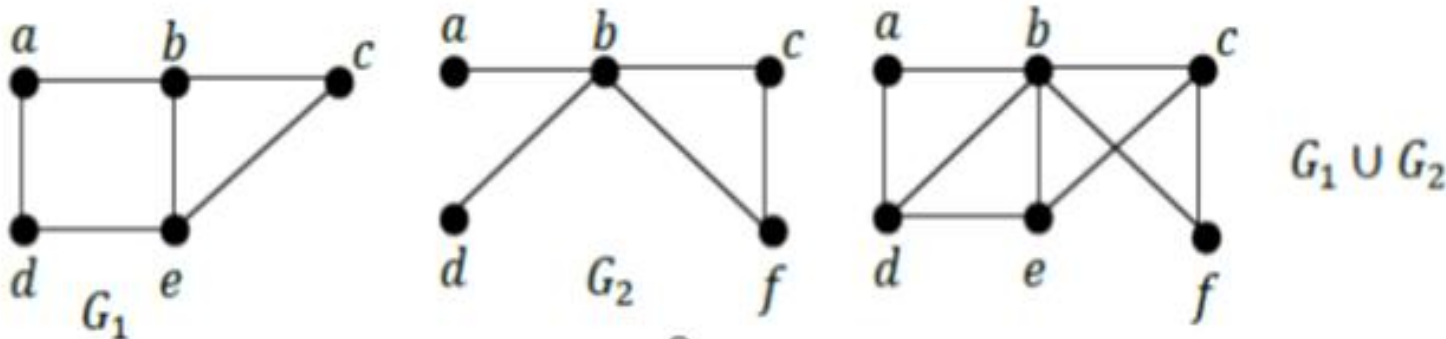
Definition: A subgraph of a graph $G = (V, E)$ is a graph $H = (W, F)$, where $W \subseteq V$ and $F \subseteq E$. A subgraph H of G is a proper subgraph of G if $H \neq G$.

For example, the graph H here is a proper subgraph of the graph G

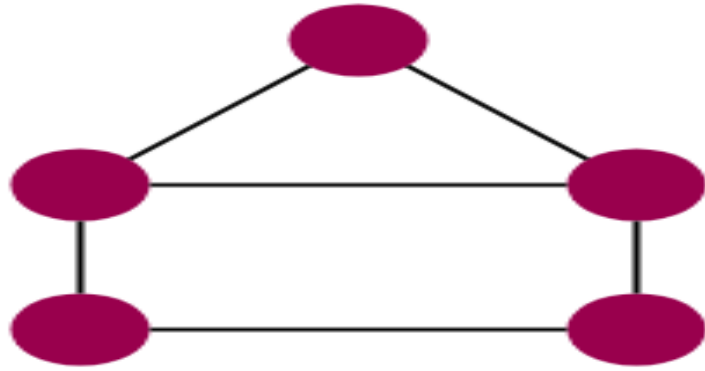


Definition: Two or more graphs can be combined in various ways. The new graph that contains all the vertices and the edges of these graphs is called the union of the graphs. Which means, the **union** of two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is the simple graph with vertex set $V_1 \cup V_2$ and the edge set $E_1 \cup E_2$. The union of G_1 and G_2 is denoted by $G_1 \cup G_2$.

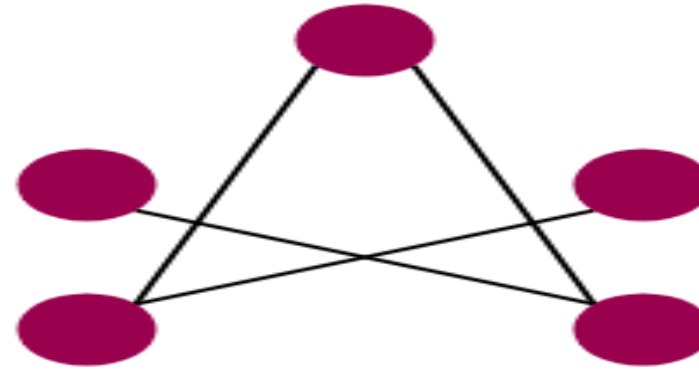
For example



Definition: The **complementary graph** $\bar{G}$ of a simple graph G has the same set of vertices as G . But two vertices are adjacent in $\bar{G}$ if they are not adjacent in G .



Graph G

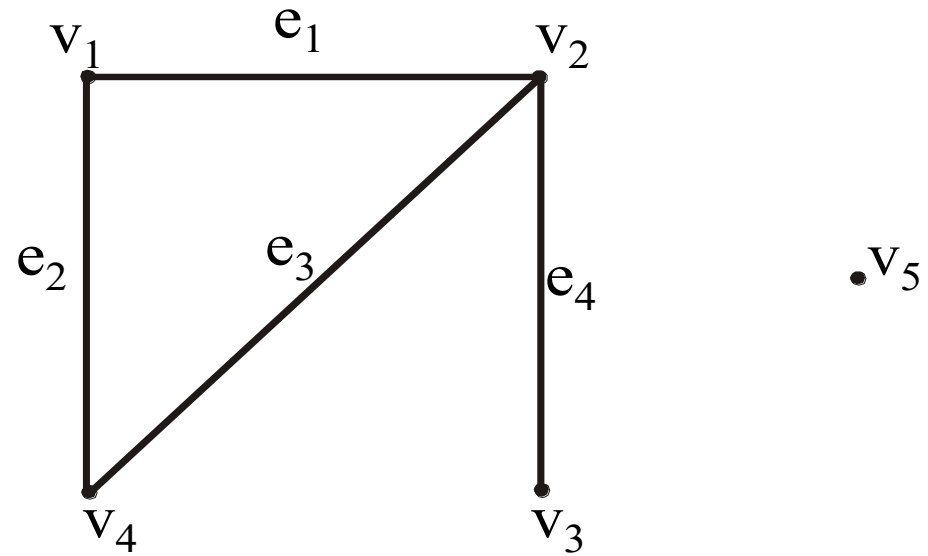


Complement Graph $\bar{G}$

4.2 Representation of Graphs and Isomorphism

Representing graphs as sets and bags: Recall that a graph consists of vertices and edges, and that edges connect vertices, we may represent a graph in terms of a set of vertices and a bag of edges. We write the graph $G = (V, E)$.

In the graph G ,
vertex set $V(G) = \{v_1, v_2, v_3, v_4, v_5\}$
edge set $E(G) = \{e_1, e_2, e_3, e_4\}$



Representing graphs as matrices:

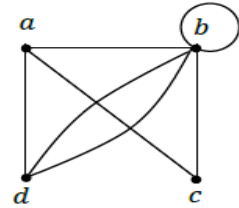
1. Adjacency Matrix

Definition: Suppose that $G = (V, E)$ is an undirected graph where $|V| = n$. Suppose that the vertices of G are listed arbitrary as $v_1, v_2, \dots, v_n$. Then the **adjacency matrix** A or A_G of the undirected graph G , with respect to this listing of vertices, is the $n \times n$ matrix where $A = [a_{ij}]$ such that each entry a_{ij} represent the number of edges between v_i and v_j .

Note: The loop in an undirected graph is counted as one edge.

Example: Find the adjacency matrix of the given graph

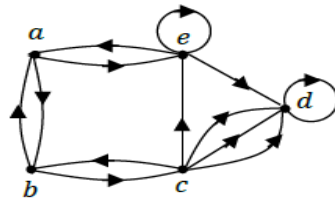
Solution: $A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \end{bmatrix}$



Definition: Suppose that $G = (V, E)$ is a directed graph where $|V| = n$. Suppose that the vertices of G are listed arbitrary as $v_1, v_2, \dots, v_n$. Then the **adjacency matrix** A or A_G of the directed graph G , with respect to this listing of vertices, is the $n \times n$ matrix where $A = [a_{ij}]$ such that each entry a_{ij} represent the number of edges started with vertex v_i and ended with the vertex v_j .

Example: Find the adjacency matrix of the given graph

Solution: $A = \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 3 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix}$



Note: The adjacency matrix of an undirected graph is a symmetric matrix while it is not necessary to be symmetric if the graph is directed.

2. Incidence Matrix

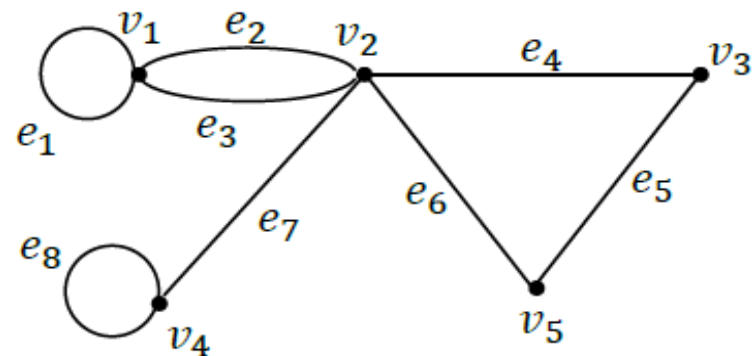
Definition: Suppose that $G = (V, E)$ is an undirected graph where $|V| = n$. Suppose that the vertices and the edges of G are listed arbitrary as $v_1, v_2, \dots, v_n$ and $e_1, e_2, \dots, e_m$ respectively. Then the **incidence matrix** M or M_G of the undirected graph G , with respect to this ordering of vertices and edges, is the $n \times m$ matrix where $M = [m_{ij}]$ where

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i, \\ 0 & \text{otherwise.} \end{cases}$$

Example: Find the incidence matrix of the given pseudograph

Solution:

$$M = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

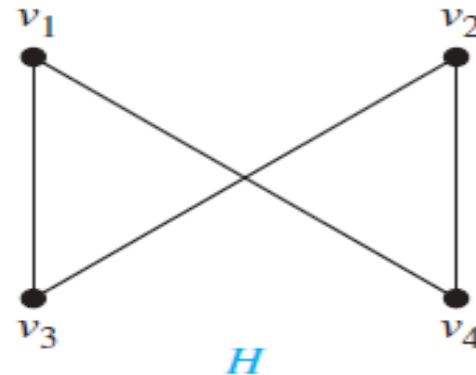
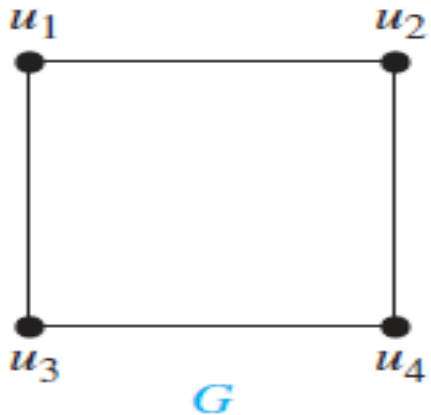


Isomorphism of Graphs:

The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are *isomorphic* if there exists a one-to-one and onto function f from V_1 to V_2 with the property that a and b are adjacent in G_1 if and only if $f(a)$ and $f(b)$ are adjacent in G_2 , for all a and b in V_1 . Such a function f is called an *isomorphism*. Two simple graphs that are not isomorphic are called *non-isomorphic*.

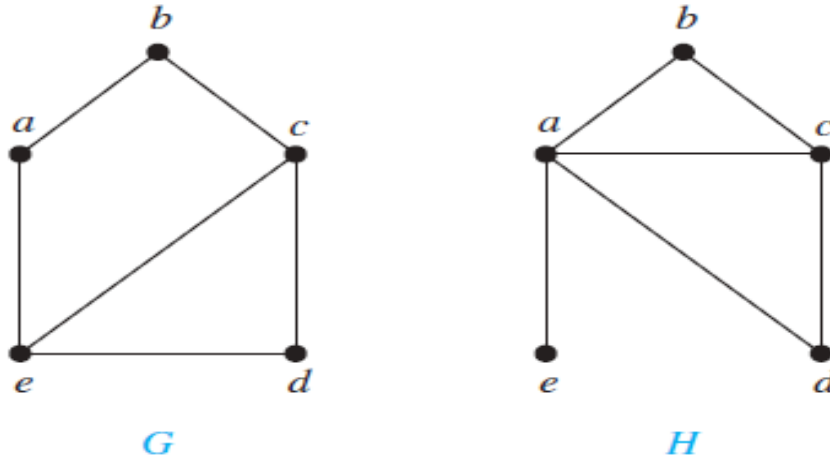
In other words, when two simple graphs are isomorphic, there is a one-to-one correspondence between vertices of the two graphs that preserves the adjacency relationship. Isomorphism of simple graphs is an equivalence relation.

Example: Show that the graphs $G = (V, E)$ and $H = (W, F)$ are isomorphic.



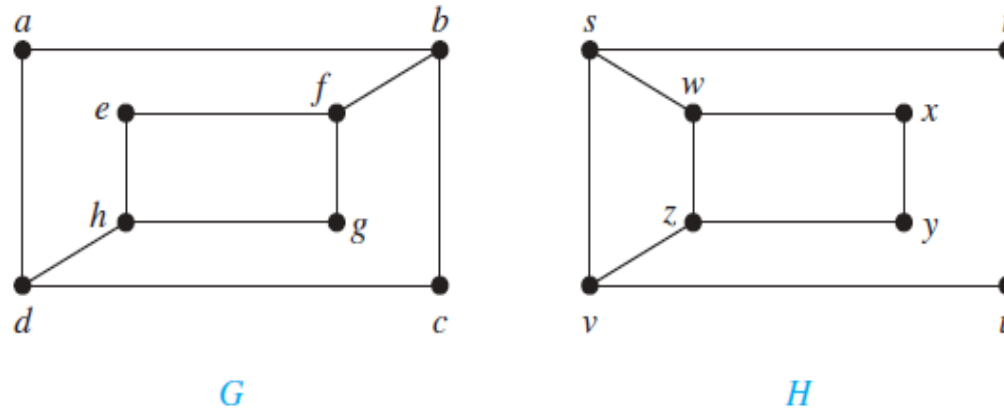
Solution: The function f with $f(u_1) = v_1, f(u_2) = v_4, f(u_3) = v_3$, and $f(u_4) = v_2$ is a one-to-one correspondence between V and W . To see that this correspondence preserves adjacency, note that adjacent vertices in G are u_1 and u_2, u_1 and u_3, u_2 and u_4 , and u_3 and u_4 , and each of the pairs $f(u_1) = v_1$ and $f(u_2) = v_4, f(u_1) = v_1$ and $f(u_3) = v_3, f(u_2) = v_4$ and $f(u_4) = v_2$, and $f(u_3) = v_3$ and $f(u_4) = v_2$ consists of two adjacent vertices in H .

Example: Show that the following graphs are not isomorphic.



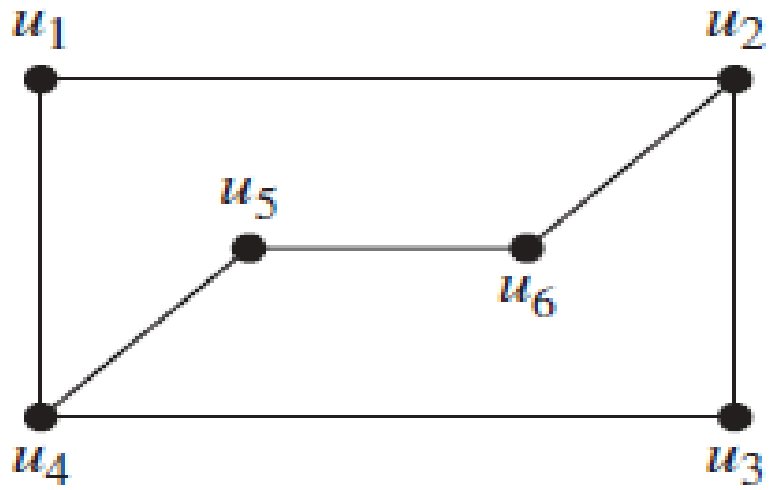
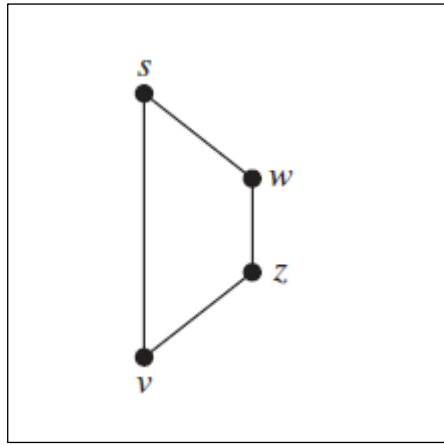
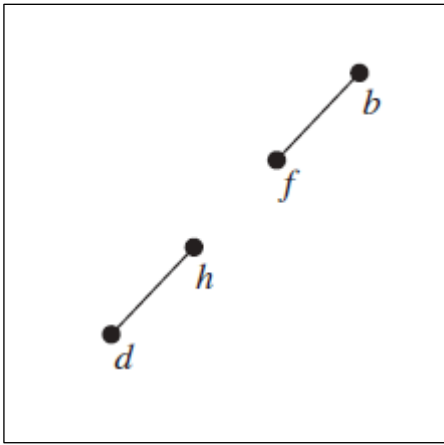
Solution: Both G and H have five vertices and six edges. However, H has a vertex of degree one, namely, e , whereas G has no vertices of degree one. It follows that G and H are not isomorphic.

Example: Determine whether the following graphs are isomorphic.

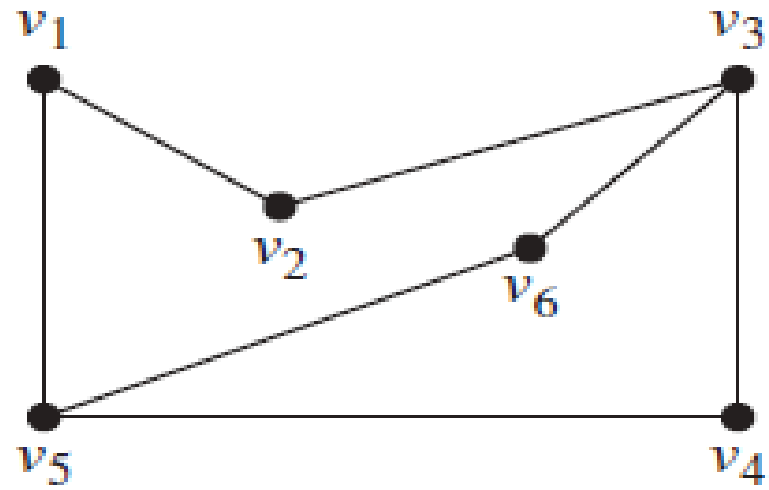


Solution: The graphs G and H both have eight vertices and 10 edges. They also both have four vertices of degree two and four of degree three. Because these invariants all agree, it is still conceivable that these graphs are isomorphic. However, G and H are not isomorphic. To see this, note that because $\deg(a) = 2$ in G , a must correspond to either t, u, x , or y in H , because these are the vertices of degree two in H . However, each of these four vertices in H is adjacent to another vertex of degree two in H , which is not true for a in G .

Another way to see that G and H are not isomorphic is to note that the subgraphs of G and H made up of vertices of degree three and the edges connecting them must be isomorphic if these two graphs are isomorphic. However, these subgraphs, shown below figure, are not isomorphic.



G



H

Solution: Both G and H have six vertices and seven edges. Both have four vertices of degree two and two vertices of degree three. It is also easy to see that the subgraphs of G and H consisting of all vertices of degree two and the edges connecting them are isomorphic (as the reader should verify). Because G and H agree with respect to these invariants, it is reasonable to try to find an isomorphism f .

We now will define a function f and then determine whether it is an isomorphism. Because $\deg(u_1) = 2$ and because u_1 is not adjacent to any other vertex of degree two, the image of u_1 must be either v_4 or v_6 , the only vertices of degree two in H not adjacent to a vertex of degree two. We arbitrarily set $f(u_1) = v_6$. [If we found that this choice did not lead to isomorphism, we would then try $f(u_1) = v_4$.] Because u_2 is adjacent to u_1 , the possible images of u_2 are v_3 and v_5 . We arbitrarily set $f(u_2) = v_3$.

Continuing in this way, using adjacency of vertices and degrees as a guide, we set $f(u_3) = v_4$, $f(u_4) = v_5$, $f(u_5) = v_1$, and $f(u_6) = v_2$. We now have a one-to-one correspondence between the vertex set of G and the vertex set of H , namely, $f(u_1) = v_6$, $f(u_2) = v_3$, $f(u_3) = v_4$, $f(u_4) = v_5$, $f(u_5) = v_1$, $f(u_6) = v_2$. To see whether f preserves edges, we examine the adjacency matrix of G ,

$$\mathbf{A}_G = \begin{matrix} & \begin{matrix} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 \end{matrix} \\ \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix},$$

and the adjacency matrix of H with the rows and columns labeled by the images of the corresponding vertices in G ,

$$\mathbf{A}_H = \begin{matrix} & \begin{matrix} v_6 & v_3 & v_4 & v_5 & v_1 & v_2 \end{matrix} \\ \begin{matrix} v_6 \\ v_3 \\ v_4 \\ v_5 \\ v_1 \\ v_2 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}.$$

Because $\mathbf{A}_G = \mathbf{A}_H$, it follows that f preserves edges. We conclude that f is an isomorphism, so G and H are isomorphic. Note that if f turned out not to be an isomorphism, we would *not* have established that G and H are not isomorphic, because another correspondence of the vertices in G and H may be an isomorphism.

4.3 Shortest-Path Problems:

In graph theory, the shortest path problem is the problem of finding a path between two vertices (or nodes) in a graph such that the sum of the weights of its constituent edges is minimized.

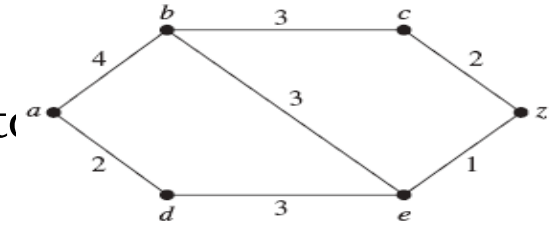
Graphs that have a number assigned to each edge are called **weighted graphs**. Weighted graphs are used to model computer networks. Communications costs (such as the monthly cost of leasing a telephone line), the response times of the computers over these lines, or the distance between computers, can all be studied using weighted graphs.

A Shortest-Path Algorithm: There are several different algorithms that find a shortest path between two vertices in a weighted graph. We will present a greedy algorithm discovered by the Dutch mathematician Edsger Dijkstra in 1959. The version we will describe solves this problem in undirected weighted graphs where all the weights are positive. It is easy to adapt it to solve shortest-path problems in directed graphs.

Before giving a formal presentation of the Dijkstra's algorithm, we will illustrate the following example.

Example: What is the length of a shortest path between a and z in the following weighted graph?

Solution: Although a shortest path is easily found by inspection, we will develop some ideas useful in understanding Dijkstra's algorithm. We will solve this problem by finding the length of a shortest path from a to z until z is reached.



A path that starts at a that has no interior vertices other than a must consist of one edge that has a as one of its endpoints. These are the one-edged paths a, b of length 4 and a, d of length 2. It follows that d is the closest vertex to a , and the shortest path from a to d has length 2.

We can find the second closest vertex by examining all paths that begin with the shortest path from a to a vertex in the set $\{a, d\}$, followed by an edge that has one endpoint in $\{a, d\}$ and its other endpoint not in this set. There are two such paths to consider, a, d, e of length 7 and a, b of length 4. Hence, the second closest vertex to a is b , and the shortest path from a to b has length 4.

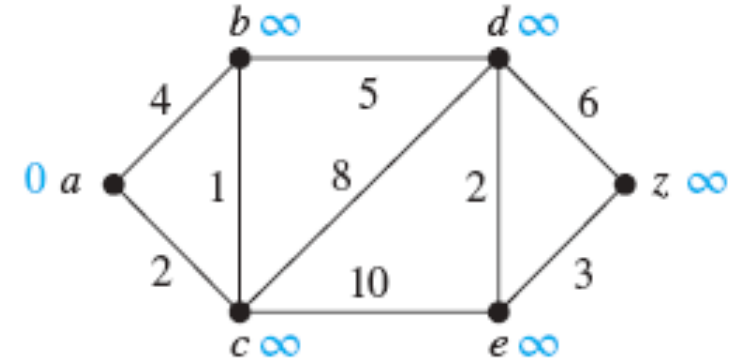
To find the third closest vertex to a , we need examine only the paths that begin with the shortest path from a to a vertex in the set $\{a, d, b\}$, followed by an edge that has one endpoint in the set $\{a, d, b\}$ and its other endpoint not in this set. There are three such paths, a, b, c of length 7, a, b, e of length 7, and a, d, e of length 5. Because the shortest of these paths is a, d, e , the third closest vertex to a is e , and the length of the shortest path from a to e is 5.

To find the fourth closest vertex to a , we need examine only the paths that begin with the shortest path from a to a vertex in the set $\{a, d, b, e\}$, followed by an edge that has one endpoint in the set $\{a, d, b, e\}$ and its other endpoint not in this set. There are two such paths, a, b, c of length 7 and a, d, e, z of length 6. Because the shorter of these paths is a, d, e, z , the fourth closest vertex to a is z , and the length of the shortest path from a to z is 6.

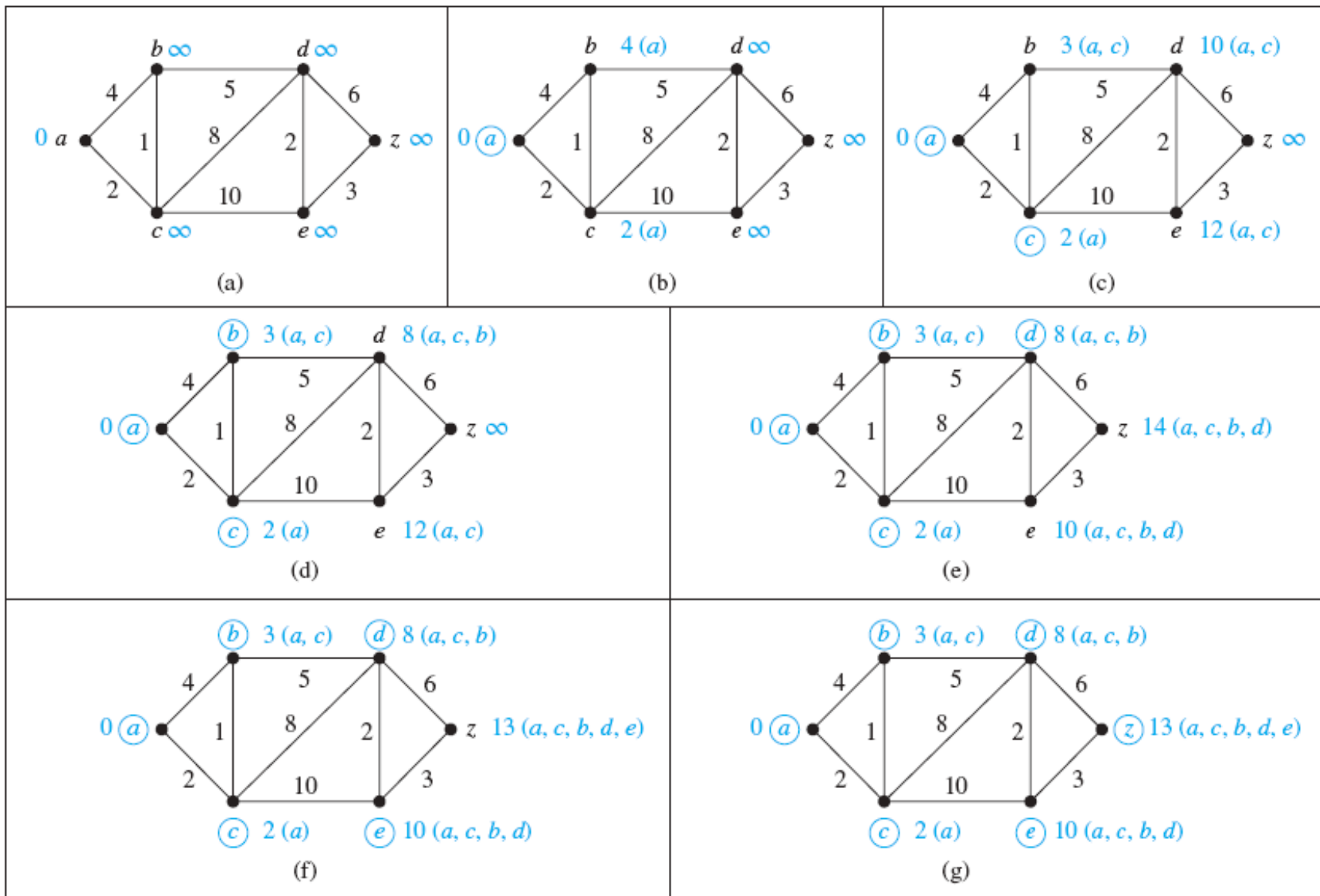
ALGORITHM 1 Dijkstra's Algorithm.

procedure *Dijkstra*(G : weighted connected simple graph, with
all weights positive)
{ G has vertices $a = v_0, v_1, \dots, v_n = z$ and lengths $w(v_i, v_j)$
where $w(v_i, v_j) = \infty$ if $\{v_i, v_j\}$ is not an edge in G }
for $i := 1$ **to** n
 $L(v_i) := \infty$
 $L(a) := 0$
 $S := \emptyset$
{ the labels are now initialized so that the label of a is 0 and all
other labels are ∞ , and S is the empty set }
while $z \notin S$
 $u :=$ a vertex not in S with $L(u)$ minimal
 $S := S \cup \{u\}$
 for all vertices v not in S
 if $L(u) + w(u, v) < L(v)$ **then** $L(v) := L(u) + w(u, v)$
 { this adds a vertex to S with minimal label and updates the
 labels of vertices not in S }
return $L(z)$ { $L(z)$ = length of a shortest path from a to z }

Example: Use Dijkstra's algorithm to find the length of a shortest path between the vertices a and z in the following weighted graph:



Solution: The steps used by Dijkstra's algorithm to find a shortest path between a and z are shown in Figure 4. At each iteration of the algorithm the vertices of the set S_k are circled. A shortest path from a to each vertex containing only vertices in S_k is indicated for each iteration. The algorithm terminates when z is circled. We find that a shortest path from a to z is a, c, b, d, e, z , with length 13.



Remark: In performing Dijkstra's algorithm it is sometimes more convenient to keep track of labels of vertices in each step using a table instead of redrawing the graph for each step.

4.4 Connectivity

When does a computer network have the property that every pair of computers can share information, if messages can be sent through one or more intermediate computers? When a graph is used to represent this computer network, where vertices represent the computers and edges represent the communication links, this question becomes: When is there always a path between two vertices in the graph?

CONNECTEDNESS and COMPONENTS

Definition: Connected Vertices

Two vertices u and v of a graph G are said to be *connected* if there exists a path from u to v .

Definition: Connected Graph

A graph G is said to be *connected* if every pair of vertices in G are connected.

Definition: Disconnected Graph

A graph which is not connected is said to be *disconnected*. A disconnected graph is the union of two or more connected subgraphs each pair of which has no vertex in common. These disjoint connected subgraphs are called the connected components of the graph.

Definition: Connected Component

A *connected component* of a graph is a connected subgraph of largest possible size.

Note: A graph H is a connected component or component of a graph G if and only if H is a subgraph of G . H is connected. No connected subgraph of G has H as a subgraph and contains vertices or edges that are not in H .

What are the connected components of the graph H shown in Figure 3?

Solution: The graph H is the union of three disjoint connected subgraphs H_1 , H_2 , and H_3 , shown in Figure 3. These three subgraphs are the connected components of H . ◀

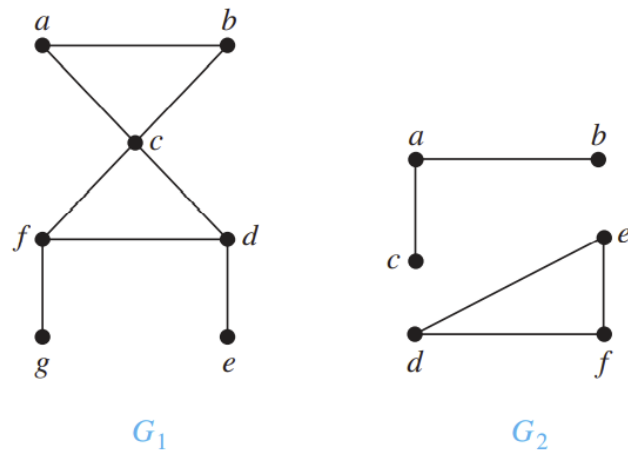


FIGURE 2 The graphs G_1 and G_2 .

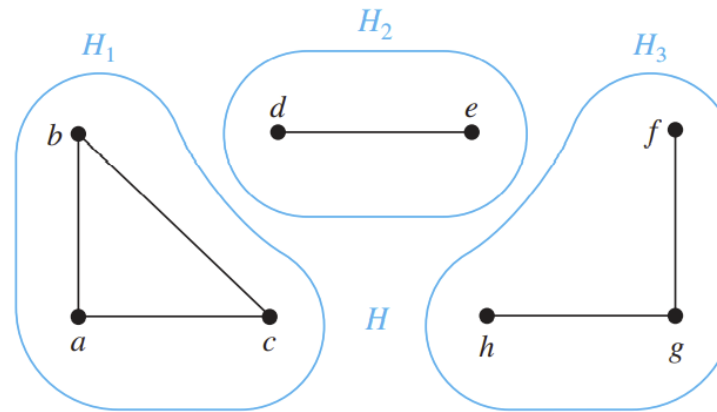


FIGURE 3 The graph H and its connected components H_1 , H_2 , and H_3 .

The graph G_1 in Figure 2 is connected, because for every pair of distinct vertices there is a path between them (the reader should verify this). However, the graph G_2 in Figure 2 is not connected. For instance, there is no path in G_2 between vertices a and d . 

Theorem: If a graph G has exactly two vertices of odd degree then there must be a path joining these vertices.

Theorem: A graph G is said to be connected iff its vertex set V can be partitioned into two subsets v_1 and v_2 such that there exists an edge in G whose one vertex is in subset of v_1 and the other is in subset of v_2 .

Theorem: If G is not connected then $\bar{G}$ is connected.

Theorem: A graph with n vertices and k components cannot have more than $\frac{(n-k)(n-k+1)}{2}$ edges.

Definition: Distance

Let u and v be any two vertices of the connected graph G . The distance between u and v is denoted by $d(u, v)$ and is defined as the length of the shortest path.

Definition: Diameter

The diameter of a connected graph G is denoted by $\text{diam}(G)$ and is defined as the maximum distance between any two vertices in G .

Definition: Cut Vertex

A cut vertex of a connected graph G is a vertex whose removal increases the number of components.

Definition: Bridge

A cut edge or bridge of a connected graph G is an edge whose removal increases the number of components.

Definition: Cut Set

The set of all minimum number of edges of G whose removal disconnects a graph G is called a cut set of G . Thus, a cut set S of a graph G satisfies the following properties:

S is a subset of the edge set E of G

Removal of edges from a connected graph G disconnects G

No proper subset of S satisfies the condition

Note: If v is cut vertex of connected graph G , then $G - v$ is disconnected. A cut vertex is also called a cut point.

Connectedness in Directed Graphs

A vertex v in a graph G is said to be *reachable* from the vertex u of G if there exists a path from u to v .

Definition: Connected or weakly connected

A directed graph is called *connected* or *weakly connected* if it is connected as an undirected graph in which each directed edge is converted to an undirected graph.

Definition: Unilaterally connected

A directed graph is said to be *unilaterally connected* if for any pair of vertices of the graph at least one of the vertices of the pair is reachable from the other vertex.

Definition: Strongly connected

A directed graph is called *strongly connected* if for any pair of vertices of the graph both the vertices of the pair are reachable from one another. In other words, a directed graph is *strongly connected* if there is a path from u to v and from v to u whenever u and v are vertices in the graph.

4.5 Euler and Hamilton Paths

Can we travel along the edges of a graph starting at a vertex and returning to it by traversing each edge of the graph exactly once? Similarly, can we travel along the edges of a graph starting at a vertex and returning to it while visiting each vertex of the graph exactly once? Although these questions seem to be similar, the first question, which asks whether a graph has an Euler circuit, can be easily answered simply by examining the degrees of the vertices of the graph, while the second question, which asks whether a graph has a Hamilton circuit, is quite difficult to solve for most graphs. In this section we will study these questions and discuss the difficulty of solving them. Although both questions have many practical applications in many different areas, both arose in old puzzles. We will learn about these old puzzles as well as modern practical applications.

Definitions:

Let G be a graph and let v and w be vertices in graph G .

1. WALK : A walk from v to w is a finite alternating sequence of adjacent vertices and edges of G . Thus a walk has the form

$$v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n$$

where the v 's represent vertices, the e 's represent edges $v_0=v$, $v_n=w$, and for all $i = 1, 2 \dots n$, v_{i-1} and v_i are endpoints of e_i .

The trivial walk from v to v consists of the single vertex v .

2. CLOSED WALK A closed walk is a walk that starts and ends at the same vertex.

3. CIRCUIT A circuit is a closed walk that does not contain a repeated edge. Thus a circuit is a walk of the form $v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n$ where $v_0 = v_n$ and all the e_i s are distinct.

4. SIMPLE CIRCUIT A simple circuit is a circuit that does not have any other repeated vertex except the first and last. Thus a simple circuit is a walk of the form $v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n$ where all the e_i s are distinct and all the v_j s are distinct except that $v_0 = v_n$

5. PATH A path from v to w is a walk from v to w that does not contain a repeated edge. Thus a path from v to w is a walk of the form

$$v = v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n = w$$

where all the e_i s are distinct (that is $e_i \neq e_k$ for any $i \neq k$).

6. SIMPLE PATH A simple path from v to w is a path that does not contain a repeated vertex. Thus a simple path is a walk of the form

$$v = v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n = w$$

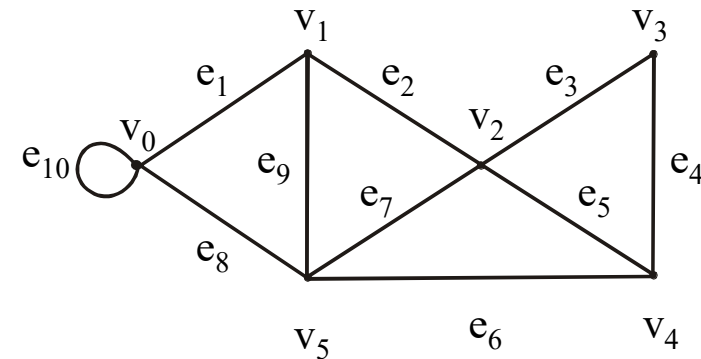
where all the e_i s are distinct and all the v_j s are also distinct (that is, $v_j \neq v_m$ for any $j \neq m$).

SUMMARY

	Repeated Edge	Repeated Vertex	Starts and Ends at Same Point
walk	allowed	Allowed	allowed
closed walk	allowed	Allowed	yes(means, where it starts also ends at that point)
circuit	no	Allowed	yes
simple circuit	no	first and last only	yes
path	no	Allowed	allowed
simple path	no	no	No

Example In the graph below, determine whether the following walks are paths, simple paths, closed walks, circuits, simple circuits, or are just walks.

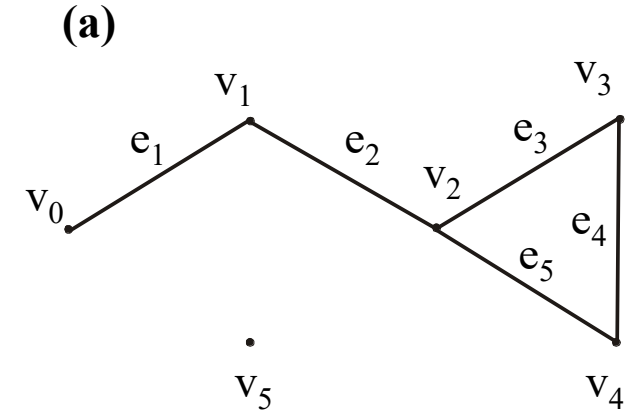
- (a) $v_1 e_2 v_2 e_3 v_3 e_4 v_4 e_5 v_2 e_2 v_1 e_1 v_0$
- (b) $v_1 v_2 v_3 v_4 v_5 v_2$
- (c) $v_4 v_2 v_3 v_4 v_5 v_2 v_4$
- (d) $v_2 v_1 v_5 v_2 v_3 v_4 v_2$
- (e) $v_0 v_5 v_2 v_3 v_4 v_2 v_1$
- (f) $v_5 v_4 v_2 v_1$



Solution:

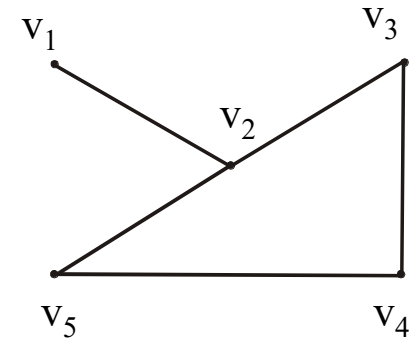
(a) $v_1 e_2 v_2 e_3 v_3 e_4 v_4 e_5 v_2 e_2 v_1 e_1 v_0$

This graph starts at vertex v_1 , then goes to v_2 along edge e_2 , and moves continuously, at the end it goes from v_1 to v_0 along e_1 . Note it that the vertex v_2 and the edge e_2 is repeated twice, and starting and ending, not at the same points. Hence The graph is just a walk.



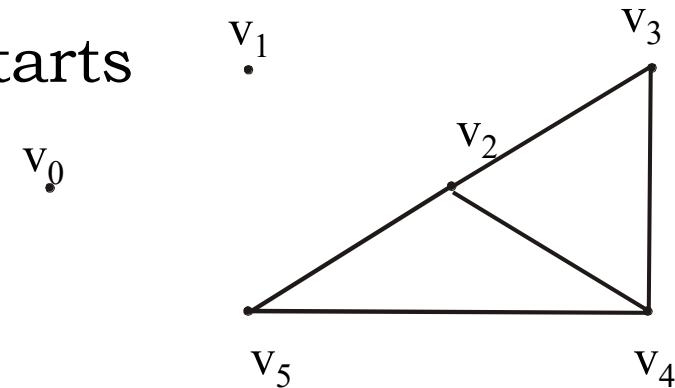
(b) $v_1 v_2 v_3 v_4 v_5 v_2$

In this graph vertex v_2 is repeated twice. As no edge is repeated so the graph is a path.



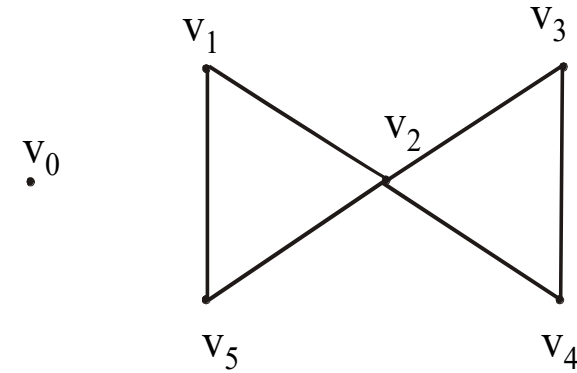
(c) $v_4 v_2 v_3 v_4 v_5 v_2 v_4$

As vertices v_2 & v_4 are repeated and graph starts and ends at the same point v_4 , also the edge (i.e. e_5) connecting v_2 & v_4 is repeated, so the graph is a closed walk.



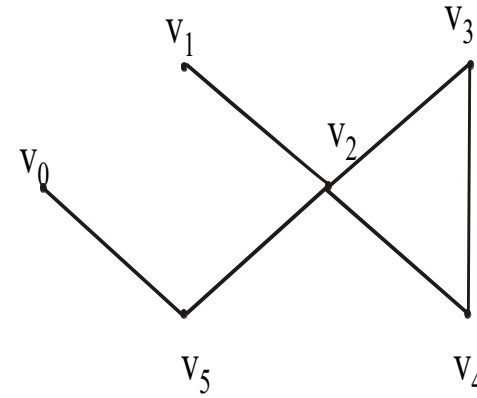
(d) $v_2v_1v_5v_2v_3v_4v_2$

In this graph, vertex v_2 is repeated and the graph starts and end at the same vertex (i.e. at v_2) and no edge is repeated, hence the above graph is a circuit.



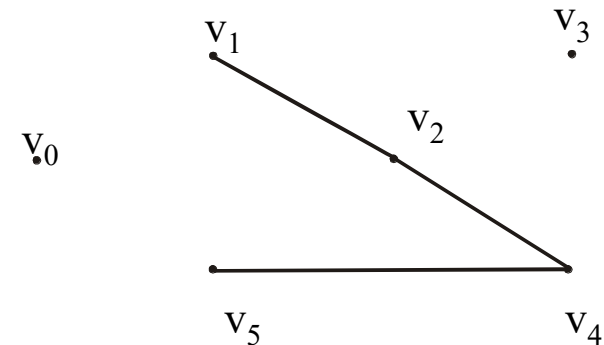
(e) $v_0v_5v_2v_3v_4v_2v_1$

Here vertex v_2 is repeated and no edge is repeated so the graph is a path.



(f) $v_5v_4v_2v_1$

Neither any vertex nor any edge is repeated so the graph is a simple path.



Euler Paths and Circuits

The town of Königsberg, Prussia (now called Kaliningrad and part of the Russian republic), was divided into four sections by the branches of the Pregel River. These four sections included the two regions on the banks of the Pregel, Kneiphof Island, and the region between the two branches of the Pregel. In the eighteenth century seven bridges connected these regions. Figure 1 depicts these regions and bridges.

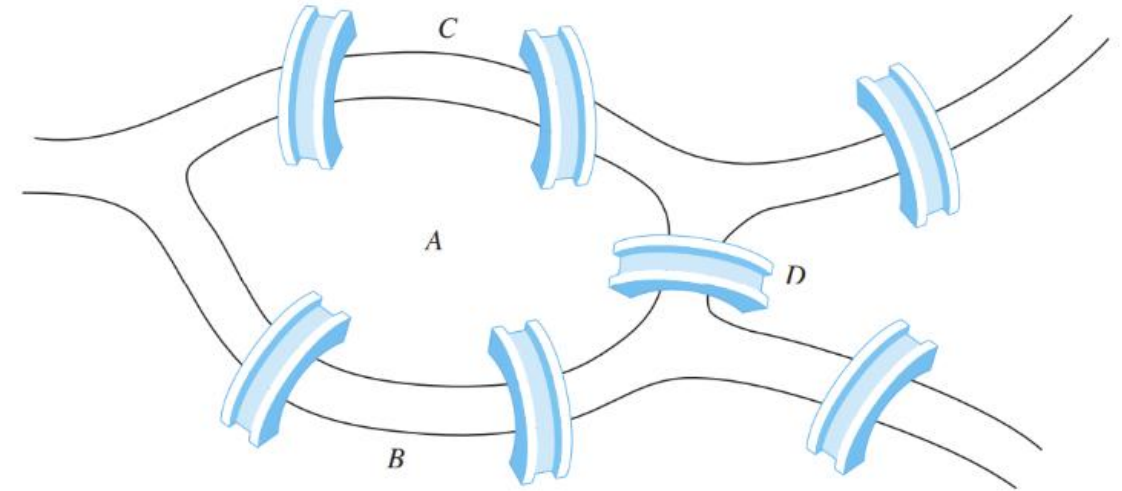


FIGURE 1 The seven bridges of Königsberg.

The townspeople took long walks through town on Sundays. They wondered whether it was possible to start at some location in the town, travel across all the bridges once without crossing any bridge twice, and return to the starting point.

The Swiss mathematician Leonhard Euler solved this problem. His solution, published in 1736, may be the first use of graph theory. (For a translation of Euler's original paper see [BiLlWi99].) Euler studied this problem using the multigraph obtained when the four regions are represented by vertices and the bridges by edges. This multigraph is shown in Figure 2. The problem of traveling across every bridge without crossing any bridge more than once can be rephrased in terms of this model. The question becomes: Is there a simple circuit in this multigraph that contains every edge?

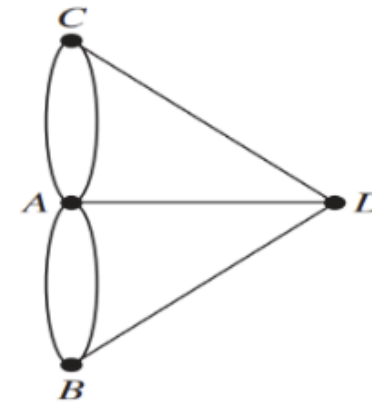


FIGURE 2 Multigraph model of the town of Königsberg.

Definition 1

An *Euler circuit* in a graph G is a simple circuit containing every edge of G . An *Euler path* in G is a simple path containing every edge of G .

Examples 1 and 2 illustrate the concept of Euler circuits and paths.

EXAMPLE 1 Which of the undirected graphs in Figure 3 have an Euler circuit? Of those that do not, which have an Euler path?

Solution: The graph G_1 has an Euler circuit, for example, a, e, c, d, e, b, a . Neither of the graphs G_2 or G_3 has an Euler circuit (the reader should verify this). However, G_3 has an Euler path, namely, a, c, d, e, b, d, a, b . G_2 does not have an Euler path (as the reader should verify). 

EXAMPLE 2 Which of the directed graphs in Figure 4 have an Euler circuit? Of those that do not, which have an Euler path?

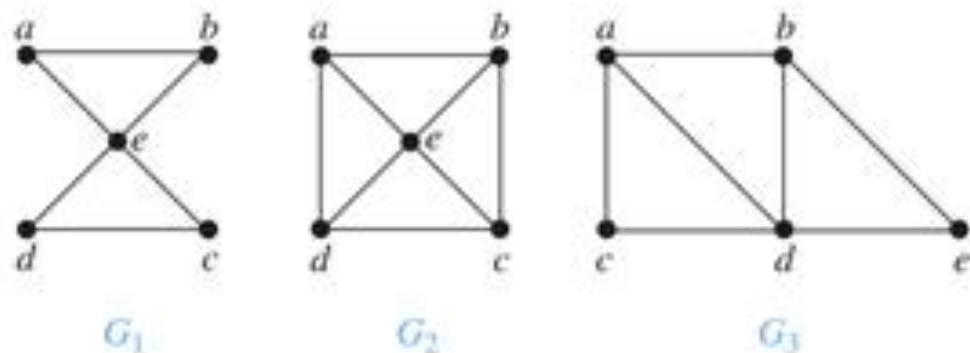


FIGURE 3 The undirected graphs G_1 , G_2 , and G_3 .

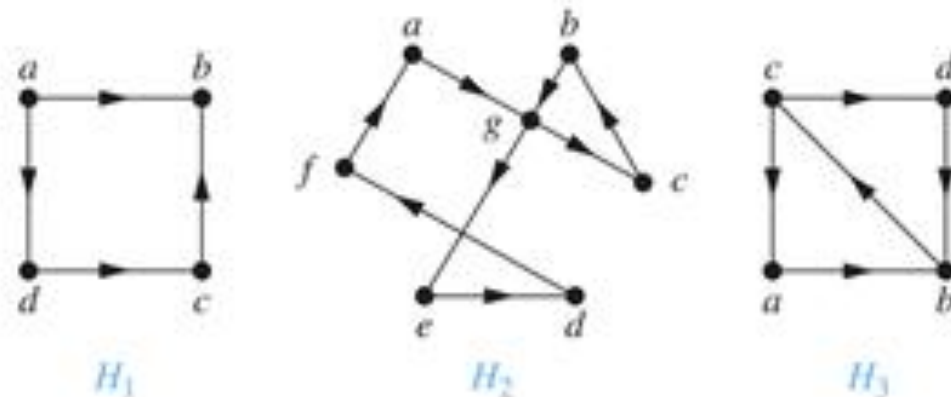


FIGURE 4 The directed graphs H_1 , H_2 , and H_3 .

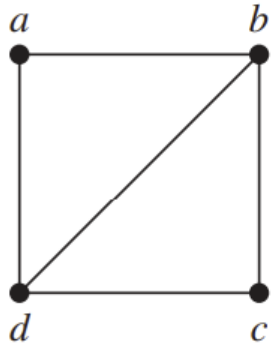
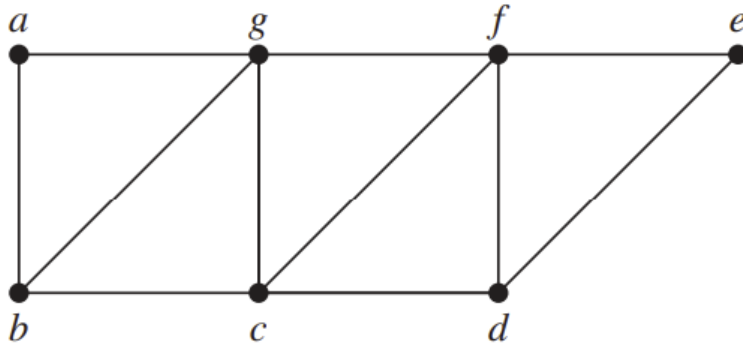
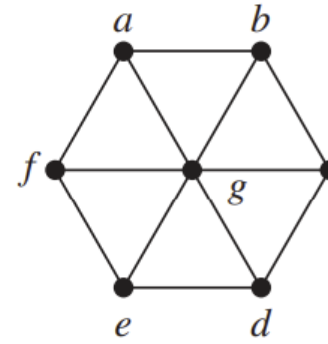
Solution: The graph H_2 has an Euler circuit, for example, $a, g, c, b, g, e, d, f, a$. Neither H_1 nor H_3 has an Euler circuit (as the reader should verify). H_3 has an Euler path, namely, c, a, b, c, d, b , but H_1 does not (as the reader should verify).

Theorems :

A connected multigraph with at least two vertices has an Euler circuit if and only if each of its vertices has even degree.

A connected multigraph has an Euler path but not an Euler circuit if and only if it has exactly two vertices of odd degree.

Which graphs shown in Figure 7 have an Euler path ?

 G_1  G_2  G_3

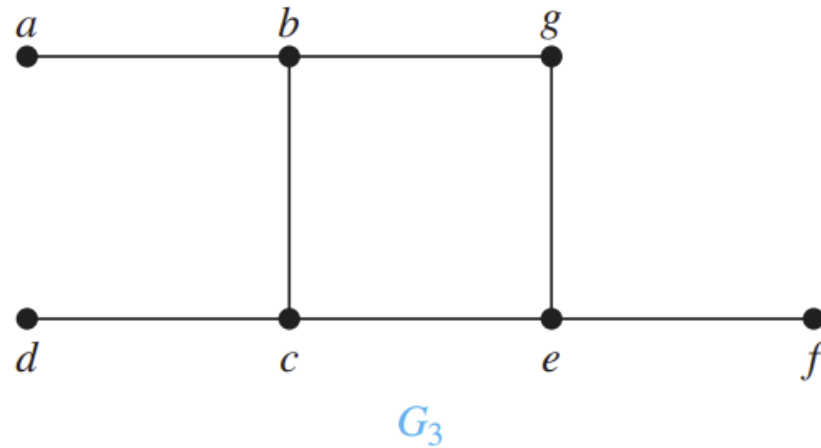
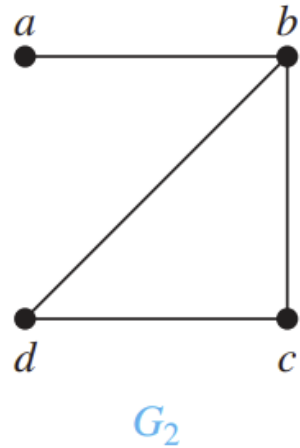
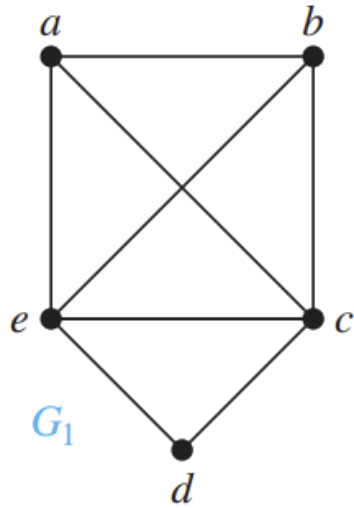
Solution: G_1 contains exactly two vertices of odd degree, namely, b and d . Hence, it has an Euler path that must have b and d as its endpoints. One such Euler path is d, a, b, c, d, b . Similarly, G_2 has exactly two vertices of odd degree, namely, b and d . So it has an Euler path that must have b and d as endpoints. One such Euler path is $b, a, g, f, e, d, c, g, b, c, f, d$. G_3 has no Euler path because it has six vertices of odd degree. 

Returning to eighteenth-century Königsberg, is it possible to start at some point in the town, travel across all the bridges, and end up at some other point in town? This question can be answered by determining whether there is an Euler path in the multigraph representing the bridges in Königsberg. Because there are four vertices of odd degree in this multigraph, there is no Euler path, so such a trip is impossible.

Hamiltonian Paths and Circuits

A simple path in a graph G that passes through every vertex exactly once is called a *Hamilton path*, and a simple circuit in a graph G that passes through every vertex exactly once is called a *Hamilton circuit*. That is, the simple path $x_0, x_1, \dots, x_{n-1}, x_n$ in the graph $G = (V, E)$ is a Hamilton path if $V = \{x_0, x_1, \dots, x_{n-1}, x_n\}$ and $x_i \neq x_j$ for $0 \leq i < j \leq n$, and the simple circuit $x_0, x_1, \dots, x_{n-1}, x_n, x_0$ (with $n > 0$) is a Hamilton circuit if $x_0, x_1, \dots, x_{n-1}, x_n$ is a Hamilton path.

EXAMPLE: Which of the simple graphs have a Hamilton circuit or, if not, a Hamilton path?



Solution: G_1 has a Hamilton circuit: a, b, c, d, e, a . There is no Hamilton circuit in G_2 (this can be seen by noting that any circuit containing every vertex must contain the edge $\{a, b\}$ twice), but G_2 does have a Hamilton path, namely, a, b, c, d . G_3 has neither a Hamilton circuit nor a Hamilton path, because any path containing all vertices must contain one of the edges $\{a, b\}$, $\{e, f\}$, and $\{c, d\}$ more than once. 

CONDITIONS FOR THE EXISTENCE OF HAMILTON CIRCUITS Is there a simple way to determine whether a graph has a Hamilton circuit or path? At first, it might seem that there should be an easy way to determine this, because there is a simple way to answer the similar question of whether a graph has an Euler circuit. Surprisingly, there are no known simple necessary and sufficient criteria for the existence of Hamilton circuits. However, many theorems are known that give sufficient conditions for the existence of Hamilton circuits. Also, certain properties can be used to show that a graph has no Hamilton circuit. For instance, a graph with a vertex of degree one cannot have a Hamilton circuit, because in a Hamilton circuit, each vertex is incident with two edges in the circuit. Moreover, if a vertex in the graph has degree two, then both edges that are incident with this vertex must be part of any Hamilton circuit. Also, note that when a Hamilton circuit is being constructed and this circuit has passed through a vertex, then all remaining edges incident with this vertex, other than the two used in the circuit, can be removed from consideration. Furthermore, a Hamilton circuit cannot contain a smaller circuit within it.

DIRAC'S THEOREM If G is a simple graph with n vertices with $n \geq 3$ such that the degree of every vertex in G is at least $n/2$, then G has a Hamilton circuit.

ORE'S THEOREM If G is a simple graph with n vertices with $n \geq 3$ such that $\deg(u) + \deg(v) \geq n$ for every pair of nonadjacent vertices u and v in G , then G has a Hamilton circuit.

4.6 Planar Graphs

Consider the problem of joining three houses to each of three separate utilities, as shown in figure 1. Is it possible to join these houses and utilities so that none of the connections cross?

This problem can be modeled using the complete bipartite graph $K_{3,3}$. The original question can be rephrased as: Can $K_{3,3}$ be drawn in the plane so that no two of its edges cross?

In this section we will study the question of whether

a graph can be drawn in the plane without edges crossing. In particular, we will answer the houses-and-utilities problem. There are always

many ways to represent a graph. When is it possible to find at least one way to represent this graph in a plane without any edges crossing?

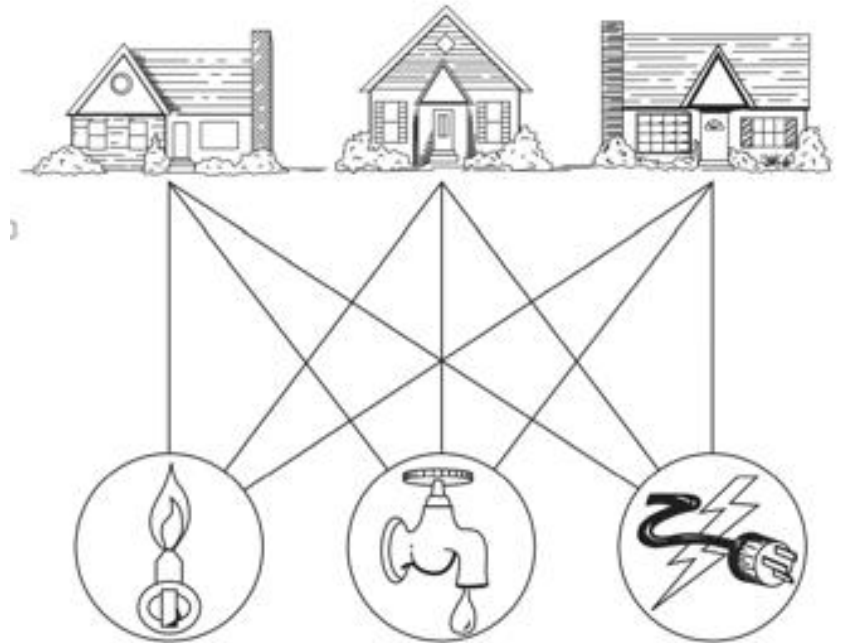


FIGURE 1 Three houses and three utilities.

A graph is called *planar* if it can be drawn in the plane without any edges crossing (where a crossing of edges is the intersection of the lines or arcs representing them at a point other than their common endpoint). Such a drawing is called a *planar representation* of the graph.

A graph may be planar even if it is usually drawn with crossings, because it may be possible to draw it in a different way without crossings.

EXAMPLE 1 Is K_4 (shown in Figure 2 with two edges crossing) planar?

Solution: K_4 is planar because it can be drawn without crossings, as shown in Figure 3.

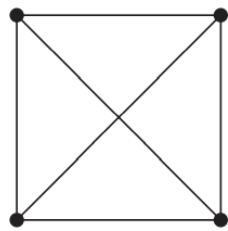


FIGURE 2 The graph K_4 .

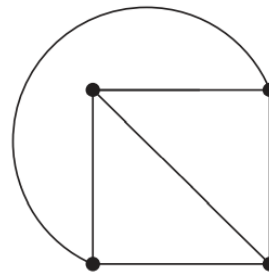


FIGURE 3 K_4 drawn with no crossings.

EXAMPLE 2 Is Q_3 , shown in Figure 4, planar?

Solution: Q_3 is planar, because it can be drawn without any edges crossing, as shown in Figure 5. ◀

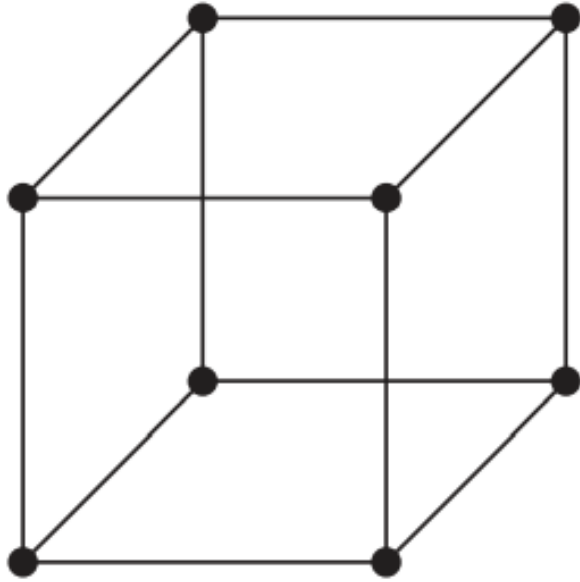


FIGURE 4 The graph Q_3 .

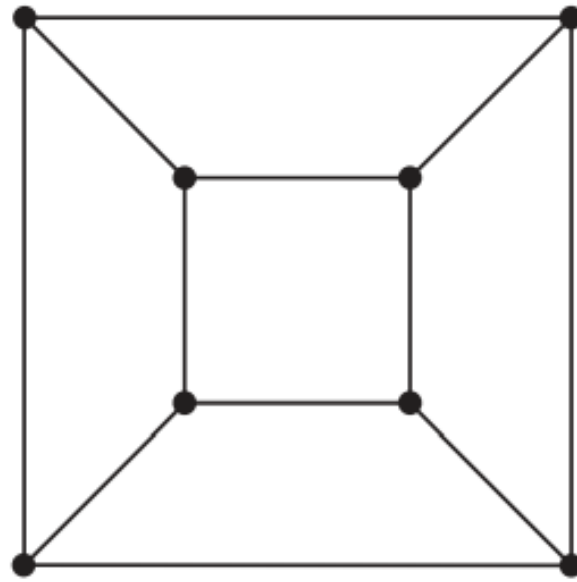


FIGURE 5 A planar representation of Q_3 .

EXAMPLE 3 Is $K_{3,3}$, shown in Figure 6, planar?

Solution: Any attempt to draw $K_{3,3}$ in the plane with no edges crossing is doomed. We now show why. In any planar representation of $K_{3,3}$, the vertices v_1 and v_2 must be connected to both v_4 and v_5 . These four edges form a closed curve that splits the plane into two regions, R_1 and R_2 , as shown in Figure 7(a). The vertex v_3 is in either R_1 or R_2 . When v_3 is in R_2 , the inside of the closed curve, the edges between v_3 and v_4 and between v_3 and v_5 separate R_2 into two subregions, R_{21} and R_{22} , as shown in Figure 7(b).

Next, note that there is no way to place the final vertex v_6 without forcing a crossing. For if v_6 is in R_1 , then the edge between v_6 and v_3 cannot be drawn without a crossing. If v_6 is in R_{21} , then the edge between v_2 and v_6 cannot be drawn without a crossing. If v_6 is in R_{22} , then the edge between v_1 and v_6 cannot be drawn without a crossing.

A similar argument can be used when v_3 is in R_1 . The completion of this argument is left for the reader (see Exercise 10). It follows that $K_{3,3}$ is not planar. ▶

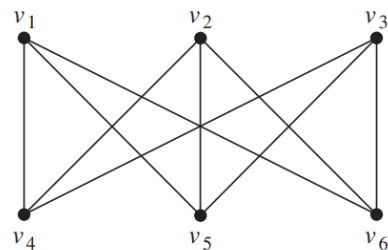


FIGURE 6 The graph $K_{3,3}$.

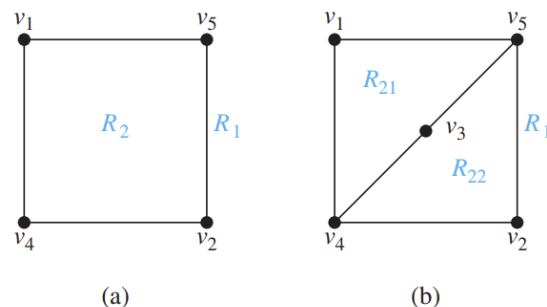


FIGURE 7 Showing that $K_{3,3}$ is nonplanar.

4.7 Graph Coloring

Problems related to the coloring of maps of regions, such as maps of parts of the world, have Links generated many results in graph theory. When a map* is colored, two regions with a common border are customarily assigned different colors. One way to ensure that two adjacent regions never have the same color is to use a different color for each region. However, this is inefficient, and on maps with many regions it would be hard to distinguish similar colors. Instead, a small number of colors should be used whenever possible. Consider the problem of determining the least number of colors that can be used to color a map so that adjacent regions never have the same color. For instance, for the map shown on the left in Figure 1, four colors suffice, but three colors are not enough. (The reader should check this.) In the map on the right in Figure 1, three colors are sufficient (but two are not). Each map in the plane can be represented by a graph. To set up this correspondence, each region of the map is represented by a vertex. Edges connect two vertices if the regions represented by these vertices have a common border. Two regions that touch at only one point are not considered adjacent. The resulting graph is called the **dual graph** of the map. By the way in which dual graphs of maps are constructed, it is clear that any map in the plane has a planar dual graph. Figure 2 displays the dual graphs that correspond to the maps shown in Figure 1. The problem of coloring the regions of a map is equivalent to the problem of coloring the vertices of the dual graph so that no two adjacent vertices in this graph have the same color. We now define a graph coloring.

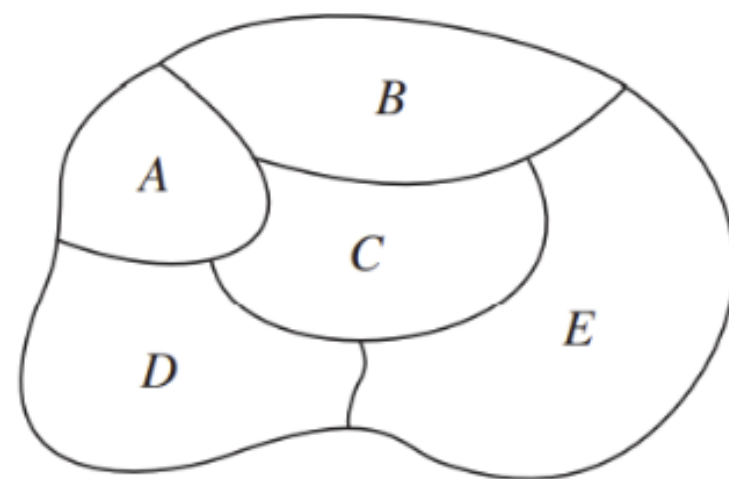
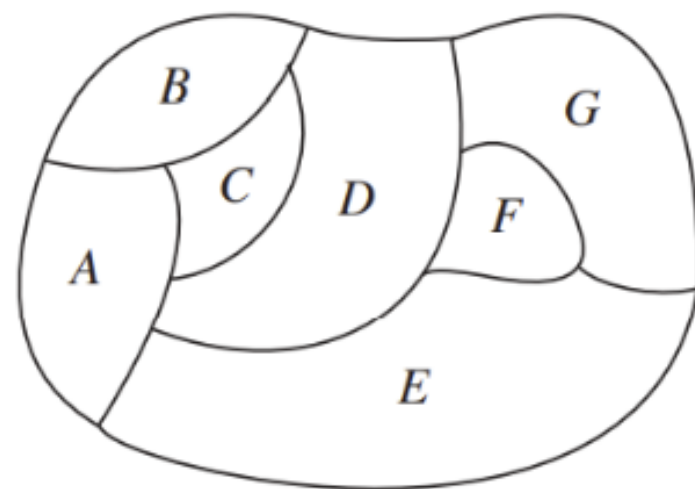


FIGURE 1 Two maps.

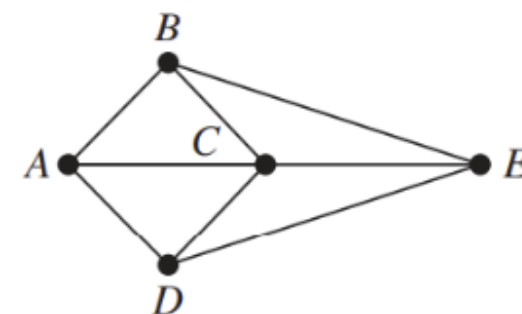
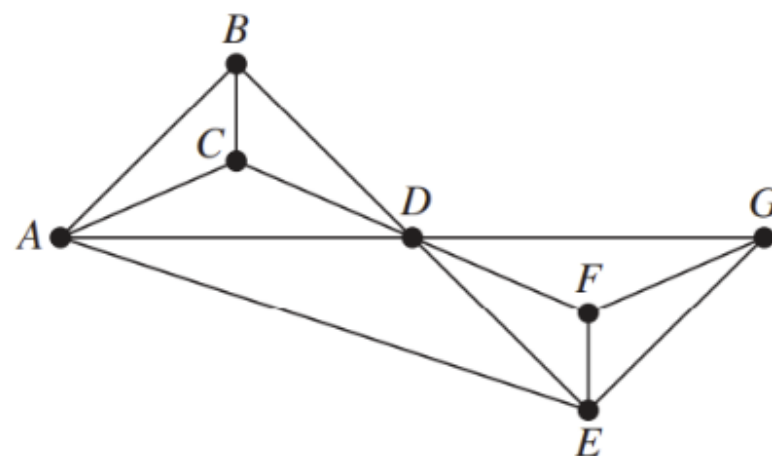


FIGURE 2 Dual graphs of the maps in Figure 1.

Definition 1

A *coloring* of a simple graph is the assignment of a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.

A graph can be colored by assigning a different color to each of its vertices. However, for most graphs a coloring can be found that uses fewer colors than the number of vertices in the graph. What is the least number of colors necessary?

Definition 2

The *chromatic number* of a graph is the least number of colors needed for a coloring of this graph. The chromatic number of a graph G is denoted by $\chi(G)$. (Here χ is the Greek letter *chi*.)

Note that asking for the chromatic number of a planar graph is the same as asking for the minimum number of colors required to color a planar map so that no two adjacent regions are assigned the same color. This question has been studied for more than 100 years. The answer is provided by one of the most famous theorems in mathematics.

THEOREM 1

THE FOUR COLOR THEOREM

The chromatic number of a planar graph is no greater than four.

EXAMPLE 1 What are the chromatic numbers of the graphs G and H shown in Figure 3?

**Extra
Examples**

Solution: The chromatic number of G is at least three, because the vertices a , b , and c must be assigned different colors. To see if G can be colored with three colors, assign red to a , blue to b , and green to c . Then, d can (and must) be colored red because it is adjacent to b and c . Furthermore, e can (and must) be colored green because it is adjacent only to vertices colored red and blue, and f can (and must) be colored blue because it is adjacent only to vertices colored red and green. Finally, g can (and must) be colored red because it is adjacent only to vertices colored blue and green. This produces a coloring of G using exactly three colors. Figure 4 displays such a coloring.

The graph H is made up of the graph G with an edge connecting a and g . Any attempt to color H using three colors must follow the same reasoning as that used to color G , except at the last stage, when all vertices other than g have been colored. Then, because g is adjacent (in H) to vertices colored red, blue, and green, a fourth color, say brown, needs to be used. Hence, H has a chromatic number equal to 4. A coloring of H is shown in Figure 4. ◀

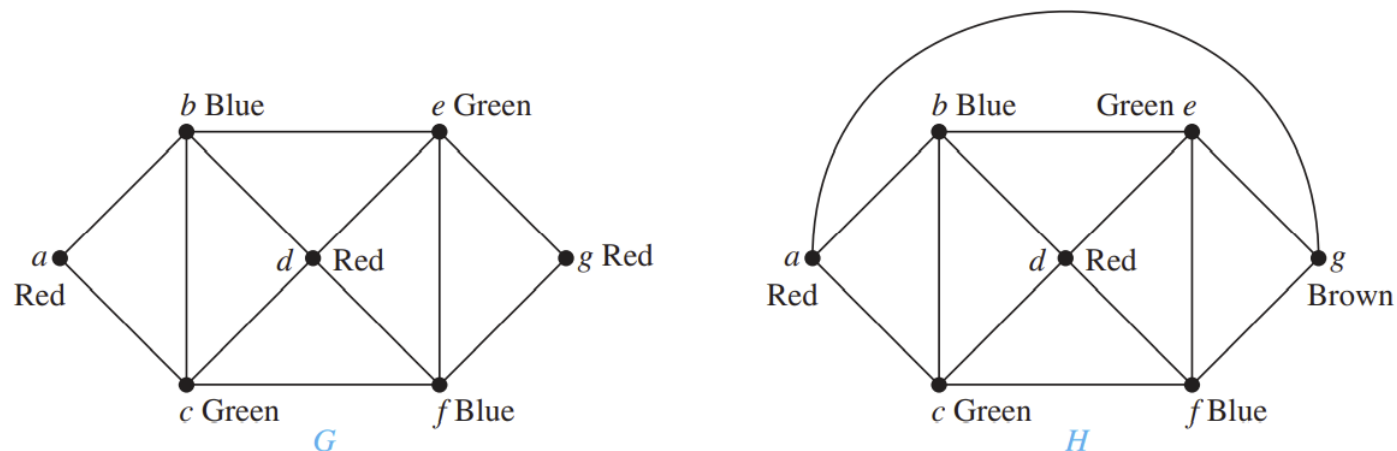


FIGURE 4 Colorings of the graphs G and H .

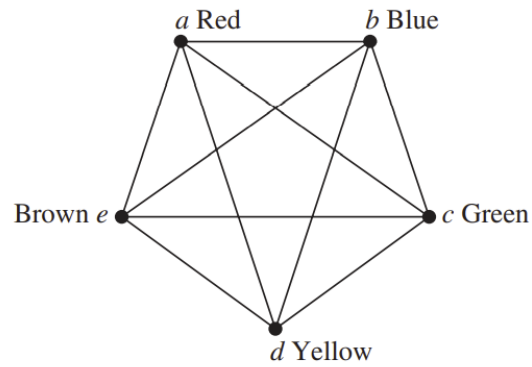


FIGURE 5 A coloring of K_5 .

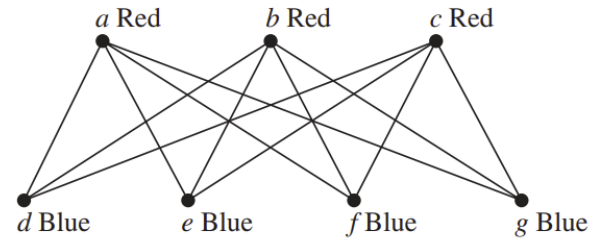


FIGURE 6 A coloring of $K_{3,4}$.

EXAMPLE 2 What is the chromatic number of K_n ?

Solution: A coloring of K_n can be constructed using n colors by assigning a different color to each vertex. Is there a coloring using fewer colors? The answer is no. No two vertices can be assigned the same color, because every two vertices of this graph are adjacent. Hence, the chromatic number of K_n is n . That is, $\chi(K_n) = n$. (Recall that K_n is not planar when $n \geq 5$, so this result does not contradict the four color theorem.) A coloring of K_5 using five colors is shown in Figure 5. ◀

EXAMPLE 3 What is the chromatic number of the complete bipartite graph $K_{m,n}$, where m and n are positive integers?

Solution: The number of colors needed may seem to depend on m and n . However, as Theorem 4 in Section 10.2 tells us, only two colors are needed, because $K_{m,n}$ is a bipartite graph. Hence, $\chi(K_{m,n}) = 2$. This means that we can color the set of m vertices with one color and the set of n vertices with a second color. Because edges connect only a vertex from the set of m vertices and a vertex from the set of n vertices, no two adjacent vertices have the same color. A coloring of $K_{3,4}$ with two colors is displayed in Figure 6. ◀